

GCP Professional Cloud Architect Crash Course

Get Fully Prepared to Crush the Exam



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Segment 1: Intro and overview of GCP exam



Objectives

- Introduction to the course
- Overview of GCP exam and what to expect

Day 1 Schedule

- **Segment 1: Intro and overview of GCP exam, what to expect**
- **Segment 2: Designing a solution infrastructure that meets business requirements**
 - Identifying business use cases and requirements
 - Success measurements
 - Planning for business continuity and cost optimization
 - Design decision trade-offs
 - Movement of data and external integrations
 - Creating a migration plan
 - Sample exam questions
- **Segment 3: Designing a solution infrastructure that meets technical requirements**
 - Overview of Google Cloud Well-Architected Framework
 - Designing for high availability and failover
 - Elasticity of cloud resources, quotas and limits
 - Designing for scalability and growth
 - Optimizing performance and latency
 - Working with Gemini Cloud Assist
 - Sample exam questions

Day 1 Schedule

- **Segment 4.1: Designing network, storage, and compute**
 - Integrations with on-premises
 - Multicloud environments
 - Cloud-native networking (VPC, peering, private service connect, etc.)
 - Choosing appropriate storage types
- **Segment 4.2: Designing network, storage, and compute**
 - Choosing data processing technologies
 - Overview of Google Cloud AI and machine learning solutions
 - Choosing compute platforms and resources
 - Sample exam questions

Standard Exam Overview

- **Length:** 2 hours
- **Registration fee:** \$200 (plus tax where applicable)
- **Languages:** English, Japanese
- **Exam format:** 50-60 multiple-choice and multiple-select questions
- **Delivery:** Remote proctored or in person at a test center.

- **Prerequisites:** None
- **Recommended experience:** 3+ years of industry experience including 1+ years with GCP
- **Certification Validity:** Two years from the date of certification.
- **Renewal:** You can choose to take a shorter (1h, 25 questions) exam.

Standard Exam Overview

Four Case Studies:

- **EHR Healthcare**
- **Cymbal Retail**
- **Altostrat Media**
- **KnightMotives**

Each exam includes 2 case studies

Case study questions make up 20-30% of the exam

Each Case Study:

- Company overview
- Solution concept
- Existing technical environment
- Business requirements
- Technical requirements
- Executive statement

<https://cloud.google.com/learn/certification/cloud-architect>

Renewal Exam Overview

- **Length:** 1 hour
- **Registration fee:** \$100 (plus tax where applicable)
- **Languages:** English, Japanese
- **Exam format:** 25 multiple-choice and multiple-select questions
- **Delivery:** Remote proctored or in person at a test center.

- **Prerequisites:** Have an active certification and within renewal eligibility period (from -60 days to +30 days from expiration)
- **Recommended experience:** 3+ years of industry experience including 1+ years with GCP
- **Certification Validity:** Two years from the date of certification.

Renewal Exam

Two Case Studies:

- **Cymbal Retail**
 - **Altostrat Media**
-
- Each exam includes 1 case study
 - Case study questions make up **90-100%** of the exam

Each Case Study:

- Company overview
- Solution concept
- Existing technical environment
- Business requirements
- Technical requirements
- Executive statement

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Tool: AWS and Azure Services Comparison

Service category ▼	Service type	Google Cloud product	Google Cloud product description	AWS offering	Azure offering
API management	API management platform	Apigee	Design, secure, analyze, and scale APIs anywhere with visibility and control.	Amazon Publisher Services, Mobile Ads	Azure API Management
API management	Monetization	Apigee API Monetization	Create new revenue streams with flexible ways to monetize your APIs.		Azure API Management
API management	Portals	Apigee integrated portals	Support for several developer portal solutions, ranging from simple turn-key solutions to solutions that are fully customizable and extensible.	Amazon API Gateway	Azure API Management
API management	API security	Advanced API Security	Help protect your APIs from security threats, including attacks from malicious clients and abuse.		Azure Defender

<https://cloud.google.com/docs/get-started/aws-azure-gcp-service-comparison>

Segment 2: Designing a solution infrastructure that meets business requirements



Objectives

- Identifying business use cases and requirements
- Success measurements
- Planning for business continuity and cost optimization
- Design decision trade-offs
- Movement of data and external integrations
- Creating a migration plan
- Sample exam questions

Business Drivers for Cloud Adoption

Why Organizations Move to Google Cloud

Digital Transformation

Modernize legacy systems, enable new business models, and accelerate innovation cycles

Scalability and Agility

Scale resources on demand to handle variable workloads without upfront capital investment

Global Reach

Deploy applications closer to users across 40+ regions worldwide

Business Outcomes to Consider

Cost Efficiency

Shift from CapEx to OpEx, pay only for resources consumed, leverage economies of scale

Security and Compliance

Meet regulatory requirements (GDPR, HIPAA, PCI-DSS) with built-in controls and certifications

AI and Analytics

Access advanced capabilities like BigQuery, Vertex AI, and Gemini models

Success Measurements

Business Metrics

Total Cost of Ownership (TCO)

Infrastructure + operations + licensing + personnel over time

Return on Investment (ROI)

Measured value delivered vs. cloud investment

Time to Market

Speed of deploying new features and applications

Customer Satisfaction

Performance impact on user experience and retention

Technical Metrics

Availability

% time operational (99.99% = 52 min downtime/year)

Latency

Response time; measured at p50, p95, p99 percentiles

Throughput

Requests/sec, transactions processed, data volume

Error Rate

% failed requests; tracked by error code category

Analyzing Scenarios and Case Studies

Extract Key Information

- **Business goals** - What outcomes does the organization need?
- **Compliance needs** - HIPAA, PCI-DSS, GDPR, industry-specific
- **Data residency** - Where must data be stored and processed?
- **Latency expectations** - Response time requirements for users
- **Growth forecasts** - Expected scale over 1-3 years
- **Integration points** - On-premises systems, third-party services

Question Framework

First Pass: Constraints

Identify non-negotiables that eliminate options (compliance, budget)

Second Pass: Requirements

Map stated needs to Google Cloud services and patterns

Third Pass: Trade-offs

Evaluate options against cost, complexity, and operational overhead

Final: Defensible Choice

Select the answer that best balances all requirements

Business Continuity

Key Definitions

Recovery Time Objective (RTO)

"How long can we be down?"

Maximum acceptable time from disruption to restoration of service

Recovery Point Objective (RPO)

"How much data can we lose?"

Maximum acceptable time since last backup or replication point

DR Pattern Cost vs. Recovery Tradeoff

Cold DR

RTO: Hours to days

RPO: Hours to days

Cost: Lowest

Backups stored in Cloud Storage; infrastructure provisioned on-demand

Warm DR

RTO: Minutes to hours

RPO: Minutes

Cost: Moderate

Scaled-down replica running; async replication; quick scale-up

Hot DR

RTO: Seconds

RPO: Near-zero

Cost: Highest

Active-active; sync replication; automatic failover

Cost Optimization Strategies

Compute Cost Reduction

Committed Use Discounts (CUDs)

1-3 year commitments for up to 57% discount on compute

Sustained Use Discounts

Automatic discounts up to 30% for VMs running 25%+ of month

Spot VMs (Preemptible)

Up to 91% discount for fault-tolerant, batch workloads

Rightsizing Recommendations

Cloud Monitoring identifies over-provisioned resources

Storage and Data

Storage Classes

Standard, Nearline, Coldline, Archive for access patterns

Object Lifecycle Management

Auto-transition or delete objects based on age

BigQuery Cost Controls

Flat-rate pricing, partitioning, clustering, materialized views

Egress Optimization

CDN caching, regional placement, Interconnect

Design Decision Trade-offs

Decision Hierarchy

1. Non-negotiable Constraints

Compliance, data residency, SLAs eliminate options immediately

2. Critical Business Requirements

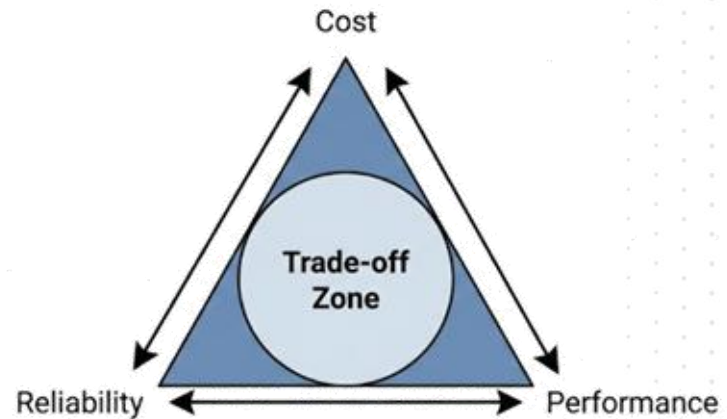
Core functionality, availability, security define solution shape

3. Optimization Goals

Cost, performance, operational simplicity refine final choice

4. Future Considerations

Scalability, extensibility, technical debt for long-term



Design Decision Trade-offs

Making Defensible Choices

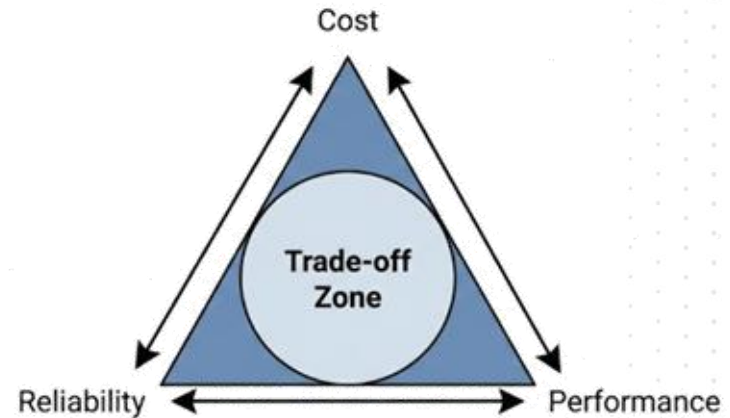
On the exam, the "best" answer balances multiple factors:

Does it meet stated requirements? Verify functional needs

Is it the simplest solution? Avoid over-engineering

Is it cost-optimal? Everything else being equal, cheaper wins

Can it be justified? Clear rationale tied to requirements



Buy vs Build vs Modify vs Deprecate

Buy (SaaS/Managed)

When: Commodity capability, not a differentiator

Examples: Cloud SQL, BigQuery, Vertex AI, Chronicle, Datadog

Pros: Faster time-to-value, less ops burden, built-in best practices

Cons: Less customization, vendor lock-in, ongoing costs

Build (Custom)

When: Unique business logic, competitive advantage

Examples: Custom ML models, proprietary monitoring solution

Pros: Full control, exact fit, IP ownership

Cons: Higher cost, longer timeline, maintenance burden

Modify (Extend)

When: You can refactor or re-architect existing apps for the cloud

Examples: Custom application monolith -> Cloud Run + Cloud SQL

Pros: Best of both: managed foundation + customization

Cons: Complexity, compatibility concerns

Deprecate (Retire)

When: Redundant, low usage, or obsolete

Examples: Legacy apps, duplicate systems, shadow IT

Pros: Reduces cost, complexity, security surface

Cons: Change management, data migration needs

Exam Tip: Default to "Buy" (managed services) unless requirements explicitly demand custom solutions.

Movement of Data: Options

Basic Transfer Methods

gsutil / gcloud storage

Command-line tools for Cloud Storage. Supports parallel composite uploads, resumable transfers. Best for ad-hoc transfers up to a few TB.

Storage Transfer Service

Managed service for large-scale transfers from AWS S3, Azure Blob, HTTP/HTTPS sources, or between GCS buckets. Supports scheduling and bandwidth limits.

BigQuery Transfer Service

Transfer data from various sources into BigQuery, optionally on a schedule.

Specialized

Transfer Appliance

Physical device shipped to your location. Rackable server (100TB-1PB). AES-256 encrypted. Ideal when network transfer would take weeks/months.

Database Migration Service

Serverless, continuous replication from MySQL, PostgreSQL, SQL Server, Oracle to Cloud SQL or AlloyDB. Minimal downtime migrations.

Datastream

Serverless CDC (change data capture) service. Streams changes from Oracle, MySQL, PostgreSQL to BigQuery, Cloud Storage, or Cloud SQL.

Movement of Data: Transferring Small Datasets

- For small transfers (<1TB and enough bandwidth), can use **gsutil** tool
 - For multi-threaded transfers, use **gsutil -m**
 - For a single large file, use **Composite transfers**
- Alternative (recommended currently): **gcloud storage**

Scenario: small (<1TB) uploads

Movement of Data: Storage Transfer Service

Transfer data securely between **object** and **file** storage across Google Cloud, AWS, Azure, and on-premises.

- Encrypts data in transit and uses checksums for data integrity checks
- Does incremental transfer
- Can set up a repeating schedule for transferring data



Scenario: less than 1TB from other clouds or on-premises

Movement of Data: BigQuery Data Transfer Service

Automates data movement into BigQuery on a schedule

- Currently can only be used to transfer data into BigQuery
- Supported sources:
 - Cloud Storage
 - Amazon S3
 - Teradata
 - Amazon Redshift
 - Google SaaS apps (Google Ads, Google Play, etc.)
 - Several third-party transfers available in Google Cloud Marketplace



Scenario: Data warehouse migration, load data into BigQuery on a schedule

Movement of Data: Storage Transfer Appliance

- High-capacity, ruggedized, tamper-resistant storage device
- Ship your data to a Google upload facility
 - Data is uploaded to **Cloud Storage**
- Data is encrypted with AES 256 encryption

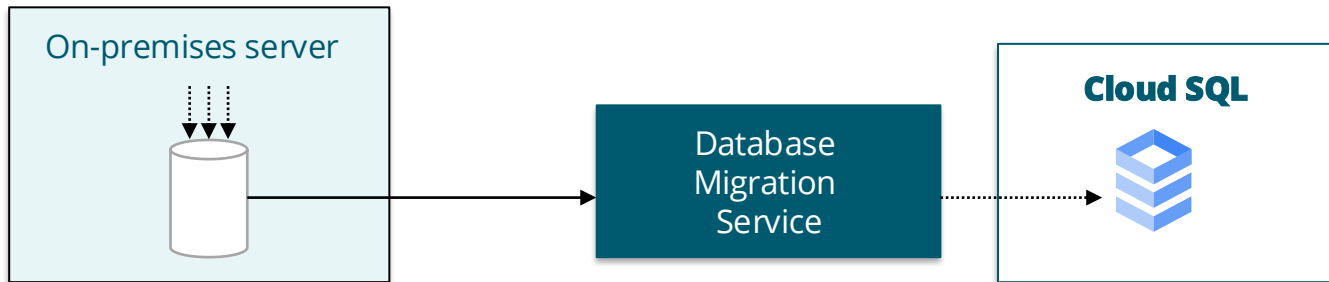


Scenario: offline very large-scale transfer

Movement of Data: Database Migration Service

Migrate databases to **Cloud SQL** or **AlloyDB**

- Serverless and guided experience
- Can migrate from on-premises, GCE, and other clouds
- Available for MySQL, PostgreSQL, Oracle, and SQL Server



Scenario: Database migration with minimal downtime

Movement of Data: Datastream

Serverless change data capture (CDC) and replication service

- Can write change event stream to Cloud Storage
- Streamlined integration with Dataflow templates to load data into multiple destinations
- Supported sources:
 - MongoDB
 - Salesforce
 - Oracle, MySQL, SQL Server, Postgres DBs



Scenario: Replicate and synchronize data across databases with minimal latency

Choosing the Right Data Transfer Method

Decision Factors

Data Volume

<1 TB: gsutil, gcloud CLI

1-10 TB: Storage Transfer Service

10+ TB: Transfer Appliance or dedicated connection

Transfer Frequency

One-time: Transfer Appliance, gsutil

Scheduled: Storage Transfer Service

Continuous: Datastream, Migration Service

Network Bandwidth

Limited/None: Transfer Appliance

Available: Online methods preferred

High-bandwidth: Interconnect + online transfer

Cost consideration: Ingress to GCP is free. Egress costs vary by destination.

Migration Strategies

Rehost

"Lift and Shift"

Move with minor or no modifications. Quickest path but no cloud optimization.

Replatform

"Lift and Optimize"

Lift then optimize for cloud. Gains elasticity, redundancy, improved security.

Refactor

"Move and Improve"

Modify to leverage cloud capabilities. Improves performance, cost, UX.

Re-architect

"Continue to Modernize"

Change how code functions. Monolith to microservices, serverless.

Rebuild

"Remove and Replace"

Completely redesign and rewrite as cloud-native. Highest effort and reward.

Repurchase

"Drop and Shop"

Replace with SaaS equivalent. Example: on-prem email to Google Workspace.

Exam Tip: Watch out for these keywords.

"Minimize changes" = **Rehost**.

"Leverage managed services" = **Replatform**.

"Cloud-native architecture" = **Re-architect** or **Rebuild**.

"Replace legacy [System]" = **Repurchase**.

Also consider **Retire** (decommission unused) and **Retain** (keep on-prem) for workloads not suited for migration.

Migration Phases

1. Assess

- Inventory workloads and dependencies
- Evaluate cloud readiness
- Estimate TCO and identify risks
- Use Migration Center for discovery

2. Plan

- Define migration waves by priority
- Establish landing zone foundation
- Set up networking and IAM
- Create runbooks and rollback plans

3. Deploy

- Execute migrations in wave
- Validate functionality at each step
- Perform cutover to production
- Decommission legacy systems

4. Optimize

- Rightsize resources based on usage
- Implement cost controls (CUDs)
- Improve automation and CI/CD
- Adopt cloud-native services

Best Practice: Start with less critical workloads to build confidence. Prioritize by business value, technical readiness, and risk tolerance.

Google Cloud Migration Tools

Assessment and Planning

Migration Center

Unified platform for discovery, assessment, and TCO analysis. Generates migration plans and Google Slides reports.

StratoZone (via Migration Center)

Deep discovery of on-premises infrastructure, dependencies, and performance data

Fit Assessment

Evaluates workload compatibility with target Google Cloud services

Migration Execution

Migrate to Virtual Machines

Rehost VMs from on-premises, AWS, or Azure to Compute Engine with minimal downtime

Migrate to Containers

Convert VM-based workloads to containers running on GKE or Cloud Run

Database Migration Service

Continuous replication for MySQL, PostgreSQL, SQL Server, Oracle to Cloud SQL or AlloyDB

Migration Best Practices



Migrate in waves

Group by dependency, risk, and priority. Start simple.

Develop runbooks

Step-by-step procedures, rollback plans, validation criteria

Enable teams

Training on GCP, operations, incident response, FinOps

Define success criteria

Performance baselines, cost targets, SLA compliance

Questions Breakdown

Your company is migrating a legacy inventory management system to Google Cloud. The application is a monolithic Java application running on VMware. The business requires the migration to be completed within 3 months with minimal disruption. The development team has limited cloud experience, and the application will eventually need to be modernized, but not during this initial migration phase.

Which migration strategy should you recommend?

- A.** Rebuild the application as a cloud-native microservices architecture on GKE
- B.** Rehost using Migrate to Virtual Machines to move VMware VMs to Compute Engine
- C.** Refactor the application to use Cloud SQL and Cloud Storage before migrating
- D.** Repurchase by replacing the system with a SaaS inventory management solution

Questions Breakdown

Your company is migrating a legacy inventory management system to Google Cloud. The application is a monolithic Java application running on VMware. The business requires the migration to be completed within 3 months with minimal disruption. The development team has limited cloud experience, and the application will eventually need to be modernized, but not during this initial migration phase.

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Questions Breakdown

A healthcare company is designing their disaster recovery strategy for a patient records system on Google Cloud. Regulations require that patient data must be recoverable within 4 hours of a disaster, and they can tolerate losing up to 1 hour of data. The system processes approximately 10,000 transactions per hour. The company wants to minimize costs while meeting these requirements.

Which disaster recovery pattern should you recommend?

A. Cold DR with daily backups to Cloud Storage and infrastructure provisioned on-demand

B. Hot DR with active-active deployment and synchronous database replication

C. Warm DR with a scaled-down replica and asynchronous replication with hourly snapshots

D. Hot DR with Cloud Spanner multi-region configuration for automatic failover

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Questions Breakdown

Your company operates a data center with 500 TB of archival data that needs to be migrated to Google Cloud Storage. The data center has a 1 Gbps internet connection shared with production workloads. The migration must be completed within 30 days, and the data is not changing during the migration window.

Which data transfer method should you use?

- A.** Use gsutil with parallel composite uploads to maximize throughput
- B.** Use Storage Transfer Service to schedule transfers during off-peak hours
- C.** Order a Transfer Appliance to physically ship the data to Google
- D.** Provision a Dedicated Interconnect connection and use Storage Transfer Service

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Questions Breakdown

You are responsible for planning a migration of your company's workloads to Google Cloud.

What actions should you do first?

- A.** Run a thorough discovery and assessment of the current environment, identify dependencies, and calculate total cost of ownership (TCO)
- B.** Set up the foundational infrastructure on GCP, decide which workload to migrate first, and run a proof-of-concept
- C.** Run a performance benchmark for your existing applications, codify the target infrastructure, and deploy a landing zone
- D.** Define the project structure on GCP, educate the team on cloud-native technologies, and modernize all applications ahead of the migration

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Questions Breakdown

Your company wants to migrate an on-premises MySQL deployment to a managed offering on Google Cloud. You need to minimize downtime and performance impact during the migration

Which approach should you recommend?

- A.** Use MySQL tools to create a dump of the existing server, then shut down the server, upload the dump file to Cloud Storage, and load it into a new Cloud SQL instance
- B.** Provision a Google Compute Engine (GCE) virtual machine and install MySQL. Set up replication on the on-premises server until cut-over
- C.** Provision a Cloud SQL for MySQL instance. Import data using the latest MySQL database backup
- D.** Provision a Cloud SQL for MySQL instance. Use the Database Migration Service to set up continuous data replication until cut-over

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Segment 3: Designing a solution infrastructure that meets technical requirements



Objectives

- Overview of Google Cloud Well-Architected Framework
- Designing for high availability and failover
- Elasticity of cloud resources, quotas and limits
- Designing for scalability and growth
- Optimizing performance and latency
- Working with Gemini Cloud Assist
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Google Cloud Well-Architected Framework

Operational Excellence

Efficient deployment, monitoring, incident response, and continuous improvement

Security

Protect data and systems with defense in depth, least privilege, zero trust

Reliability

Fault tolerance, redundancy, graceful degradation, automated recovery

Performance

Right-size resources, optimize for workload patterns, leverage caching

Cost Optimization

Align spending with value, optimize continuously, foster cost awareness

Sustainability

Reduce environmental impact through efficient resource usage

Operational Excellence

Design, implement, and manage cloud solutions that deliver value, performance, security, and reliability through **automation** and **continuous improvement**.

Infrastructure as Code

Use Terraform, Infrastructure Manager, or Pulumi for consistent, repeatable deployments. Version control all infrastructure definitions.

Observability

Implement Cloud Monitoring, Cloud Logging, and Cloud Trace. Create dashboards and alerts for proactive issue detection.

Incident Management

Define runbooks and playbooks. Use Error Reporting and integrate with PagerDuty or Opsgenie for on-call rotations.

CI/CD Pipelines

Automate builds with Cloud Build. Use Artifact Registry for container images. Implement progressive rollouts with Cloud Deploy.

Change Management

Follow safe deployment practices. Use canary and blue-green deployments. Implement feature flags for controlled rollouts.

Capacity Planning

Monitor resource utilization trends. Use recommenders for right-sizing. Plan for growth with quota management.

SRE Alignment

Follows Google SRE practices: SLIs, SLOs, error budgets

DORA Metrics

Track deployment frequency, lead time, MTTR, change failure rate

FinOps Integration

Align operational decisions with cost optimization goals

Security, Privacy, and Compliance Pillar

Identity and Access Management

- Apply principle of least privilege with IAM roles
- Use service accounts with minimal permissions
- Implement Workload Identity for GKE
- Enable MFA and use Identity-Aware Proxy (IAP)

Data Protection

- Encrypt data at rest (default) and in transit
- Use Customer-Managed Encryption Keys (CMEK)
- Implement DLP API for sensitive data discovery
- Use VPC Service Controls for data exfiltration prevention

Network Security

- Use private GKE clusters and Private Google Access
- Implement Cloud Armor for DDoS protection
- Configure firewall rules with least privilege

Security Operations

- Enable Security Command Center (SCC) Premium
- Use Chronicle for security analytics (SIEM/SOAR)
- Monitor with Cloud Audit Logs
- Implement Binary Authorization for containers

Supply Chain Security

- Use Artifact Registry with vulnerability scanning
- Implement SLSA framework for build integrity
- Sign container images with Binary Authorization
- Use Software Delivery Shield for end-to-end security

Compliance Frameworks

Google Cloud supports: SOC 1/2/3, ISO 27001, HIPAA, PCI DSS, FedRAMP, GDPR, and region-specific certifications. Use Assured Workloads for compliance-focused deployments.

Reliability Pillar

Fault Tolerance

- Design for failure with redundancy
- Use regional resources over zonal
- Implement health checks and auto-healing
- Use managed services for built-in HA

Disaster Recovery

- Define RTO and RPO requirements and test recovery procedure regularly
- Implement backup strategies (snapshots, exports)
- Use multi-region for critical workloads

Graceful Degradation

- Implement circuit breakers
- Use timeouts and retries with backoff
- Design for partial failures
- Provide fallback responses

Load Management

- Use Cloud Load Balancing for distribution
- Implement rate limiting and throttling
- Queue work with Pub/Sub or Cloud Tasks
- Cache frequently accessed data

Testing Reliability

- Conduct chaos engineering experiments
- Perform load and stress testing
- Run disaster recovery drills
- Test failover mechanisms

Observability

- Monitor SLIs aligned to user experience
- Set meaningful SLOs with error budgets
- Create actionable alerts (not noisy)
- Track dependencies and bottlenecks

99.9% SLA

8.76 hrs/year

downtime allowed

99.95% SLA

4.38 hrs/year

downtime allowed

99.99% SLA

52.6 min/year

downtime allowed

99.999% SLA

5.26 min/year

downtime allowed

Cost Optimization Pillar

Right-Sizing Resources

- Use Recommender for VM and disk sizing
- Implement custom machine types
- Monitor and eliminate idle resources
- Use autoscaling to match demand

Pricing Models

- Committed Use Discounts (CUDs) for steady workloads
- Spot VMs for fault-tolerant batch jobs (up to 91% off)
- Sustained Use Discounts (automatic)
- Flat-rate pricing for BigQuery slots

Storage Optimization

- Use appropriate storage classes (Standard, Nearline, Coldline, Archive)
- Implement Object Lifecycle Management policies
- Use Autoclass for automatic tiering
- Delete unused snapshots and images

Governance and Visibility

- Set up billing accounts with budgets and alerts
- Use labels for cost attribution
- Export billing data to BigQuery for analysis
- Implement FinOps practices with cross-functional teams

Architecture Choices

- Prefer managed/serverless services (reduce ops overhead)
- Use preemptible workers for Dataproc clusters
- Choose regional over multi-regional when appropriate
- Optimize network egress with CDN and caching

Performance Optimization Pillar

Compute Optimization

- Select appropriate machine families (E2, N2, C2, M2)
- Use GPUs/TPUs for ML workloads
- Consider serverless for variable loads
- Enable Compute Optimizer recommendations

Storage Performance

- Choose SSD vs HDD based on IOPS needs
- Use local SSD for ultra-low latency
- Optimize Cloud Storage with parallel uploads
- Use Filestore for shared file systems

Database Tuning

- Select right database for workload (SQL vs NoSQL)
- Use read replicas for read-heavy workloads
- Implement connection pooling
- Optimize queries and indexes

Network Performance

- Use Premium Tier for lowest latency
- Deploy in regions close to users
- Use Cloud CDN for static content
- Optimize with Cloud Interconnect for hybrid

Caching Strategies

- Use Memorystore (Redis/Memcached)
- Implement application-level caching
- Cache API responses with Cloud CDN
- Use BigQuery BI Engine for dashboards

Application Design

- Use async processing with Pub/Sub
- Implement batch operations
- Compress data in transit
- Profile and optimize hot paths

Cloud Trace

Distributed tracing for latency analysis

Cloud Profiler

Continuous CPU and memory profiling

Performance Dashboard

Real-time system performance metrics

Recommender

AI-powered optimization suggestions

Sustainability Pillar

Sustainable Architecture Choices

- Use serverless services that scale to zero
- Right-size resources to eliminate waste
- Delete unused resources and data
- Choose low-carbon regions when possible

Workload Optimization

- Schedule batch jobs during low-carbon periods
- Use Spot VMs to utilize spare capacity
- Minimize data movement across regions

Monitoring and Reporting

- Track carbon footprint with Carbon Footprint Dashboard
- Set sustainability goals and measure progress
- Include carbon metrics in architecture decisions
- Report on gross and net emissions

Google Cloud Commitment

Matches 100% of electricity with renewable energy.
Carbon neutral since 2007.
Target: 24/7 carbon-free energy by 2030.

100%

Renewable Energy Matched

1.1

Average PUE (Power Usage Effectiveness)

2030

24/7 Carbon-Free Target

High Availability Design

Zonal Resources

Single zone deployment. Vulnerable to zone failures.

- Compute Engine instances
- Persistent Disks
- Cloud SQL (single zone)

Regional Resources

Replicated across zones. Survives zone failures.

- Regional Managed Instance Groups
- Regional Persistent Disks
- Cloud SQL HA (automatic failover)
- Cloud Spanner (regional)

Multi-Regional Resources

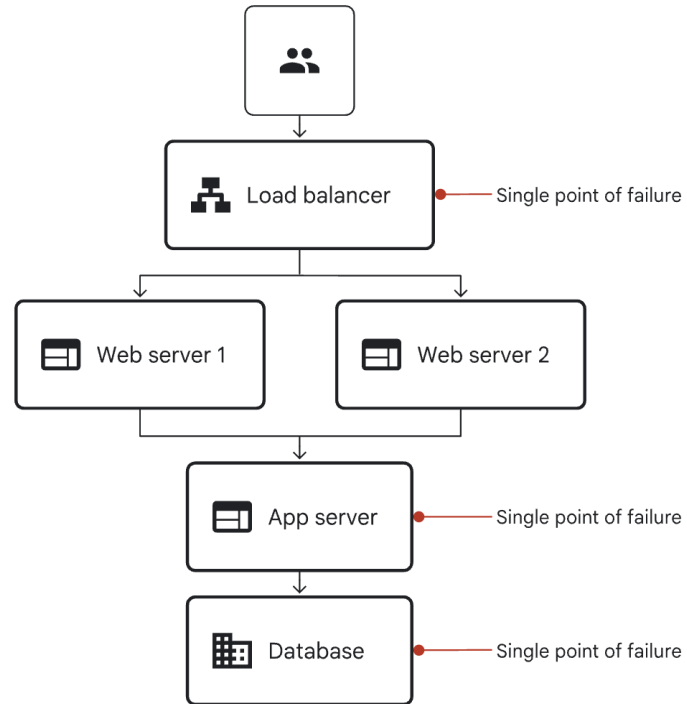
Replicated across regions. Survives region failures.

- Cloud Spanner (multi-regional)
- Cloud Storage (multi-regional)
- Firestore (multi-region)
- Global Load Balancers

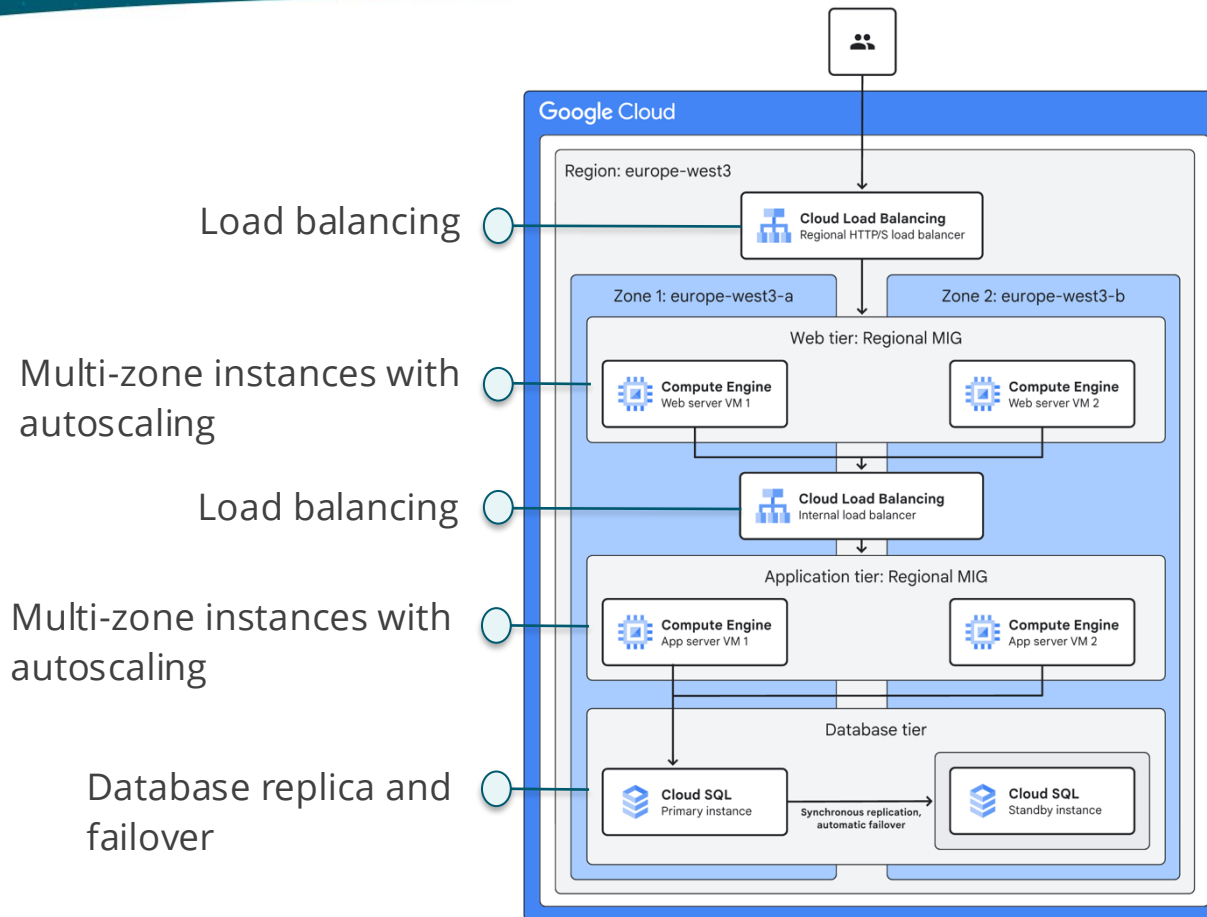
Exam Tip: Understand the trade-offs between availability levels, cost, and latency.
Higher availability typically means higher cost and potentially higher write latency due to synchronous replication.

High Availability

Avoid single point of failures!



Multi-zone HA Architecture



Elasticity

- Does the solution scale when busy, so that the service's availability and performance remain intact?
- Does the solution scale back when demand is low, so that infrastructure cost is reduced?



Elasticity and Auto-Scaling

Horizontal Scaling (Scale Out/In)

Add or remove instances based on demand. Preferred for cloud-native apps.

- Managed Instance Groups with autoscaler
- GKE Horizontal Pod Autoscaler (HPA)
- Cloud Run automatic scaling (0 to N)
- Dataproc autoscaling for Spark clusters

Vertical Scaling (Scale Up/Down)

Change instance size. Requires downtime for VMs.

- GKE Vertical Pod Autoscaler (VPA)
- Machine type changes (with restart)
- Cloud SQL storage autoscaling

Predictive Autoscaling

Uses ML to forecast load and scale proactively before demand hits.

- Analyzes historical CPU patterns
- Scales ahead of anticipated load spikes
- Reduces cold-start latency
- Available for Compute Engine MIGs

Autoscaling Metrics

- CPU utilization (default)
- Memory utilization
- Load balancing capacity
- Custom Cloud Monitoring metrics
- Pub/Sub queue depth

Quotas and Limits

Types of Quotas

- **Allocation Quotas:** Limit on resource quantity (e.g., VMs, IPs)
- **Rate Quotas:** Limit on API calls per time period
- **Concurrent Quotas:** Limit on simultaneous operations

Quota Levels

- **Project-level:** Per-project limits (most common)
- **Folder-level:** Aggregate across child projects
- **Organization-level:** Org-wide limits
- **Region/Zone:** Location-specific limits

Best Practices

- Set up quota alerts in Cloud Monitoring
- Use exponential backoff for rate limit errors
- Request increases before launch/scaling events
- Spread resources across projects if needed
- Document quota dependencies in runbooks

Requesting Quota Increases

- Use Cloud Console Quotas page
- Some quotas auto-increase with usage
- Enterprise support can expedite requests
- Plan ahead - increases can take days

Important: Limits (vs quotas) are hard caps that cannot be increased. Always check documentation to distinguish between adjustable quotas and fixed limits.

Scalability Patterns

Stateless Services

Store no session state locally. Easy to scale horizontally.

- Store sessions in Memorystore or Firestore
- Use JWTs for authentication
- Any instance can handle any request
- Ideal for Cloud Run, GKE, MIGs

Loose Coupling

Components communicate via APIs and queues, not direct calls.

- Use Pub/Sub for async messaging
- Cloud Tasks for guaranteed delivery
- Eventarc for event-driven architecture
- Scale components independently

Database Scaling

Scale data tier to match application growth.

- Read replicas for read-heavy loads
- Sharding for write-heavy loads
- Cloud Spanner for global scale
- Bigtable for high-throughput NoSQL

Caching Layer

Reduce load on backends with caching.

- Cloud CDN for static assets
- Memorystore for session/app cache
- BigQuery BI Engine for analytics

Queue-Based Processing

Buffer work to handle traffic spikes.

- Pub/Sub for message buffering
- Cloud Tasks for rate limiting
- Dataflow for stream processing

Microservices

Break monolith into independently scalable services.

- Scale bottleneck services only
- Use Anthos Service Mesh
- Independent deployment cycles

Serverless

Cloud Run, Functions: automatic scaling with zero config

Containers

GKE: HPA, VPA, and cluster autoscaler

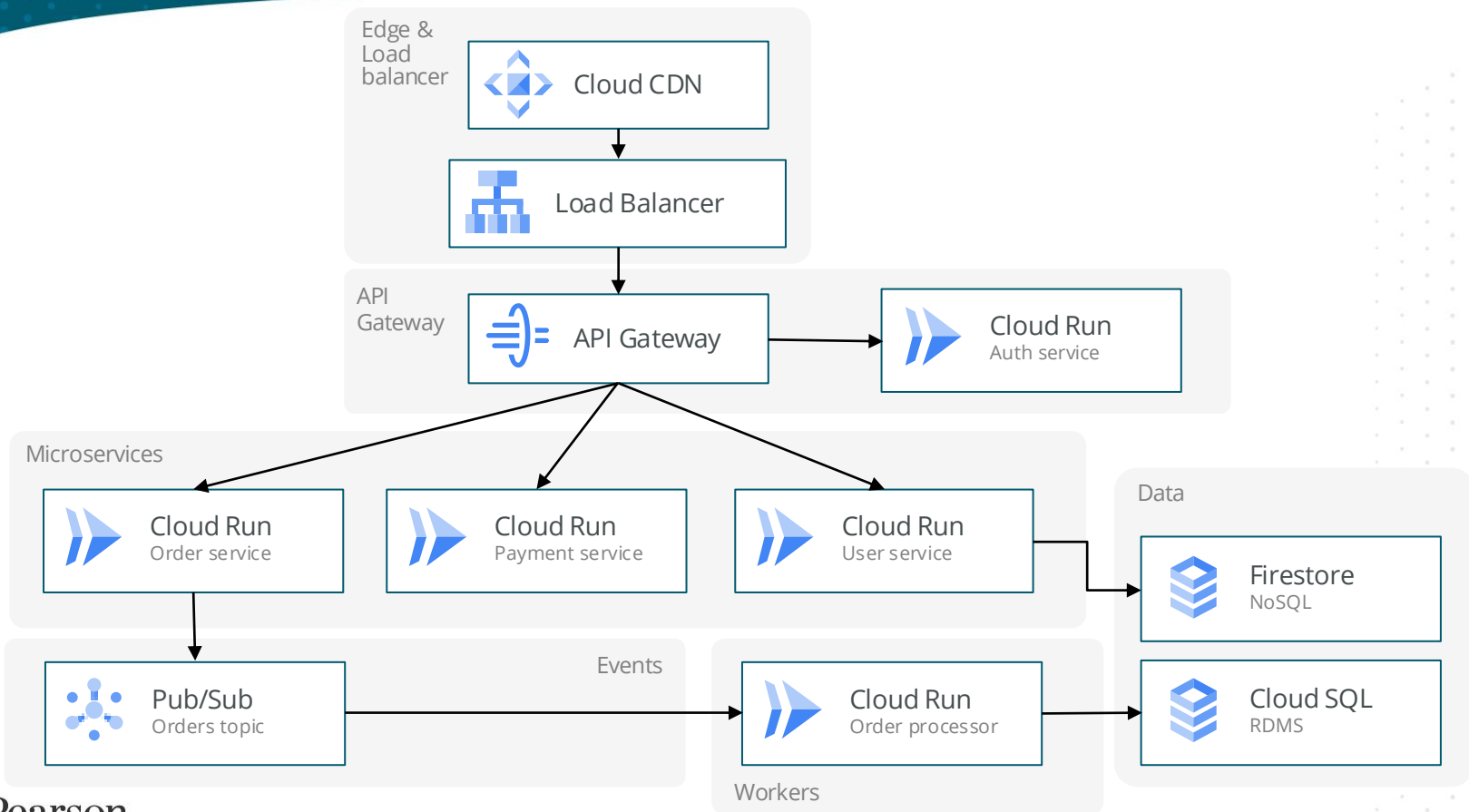
VMs

MIGs: autoscaling with custom metrics

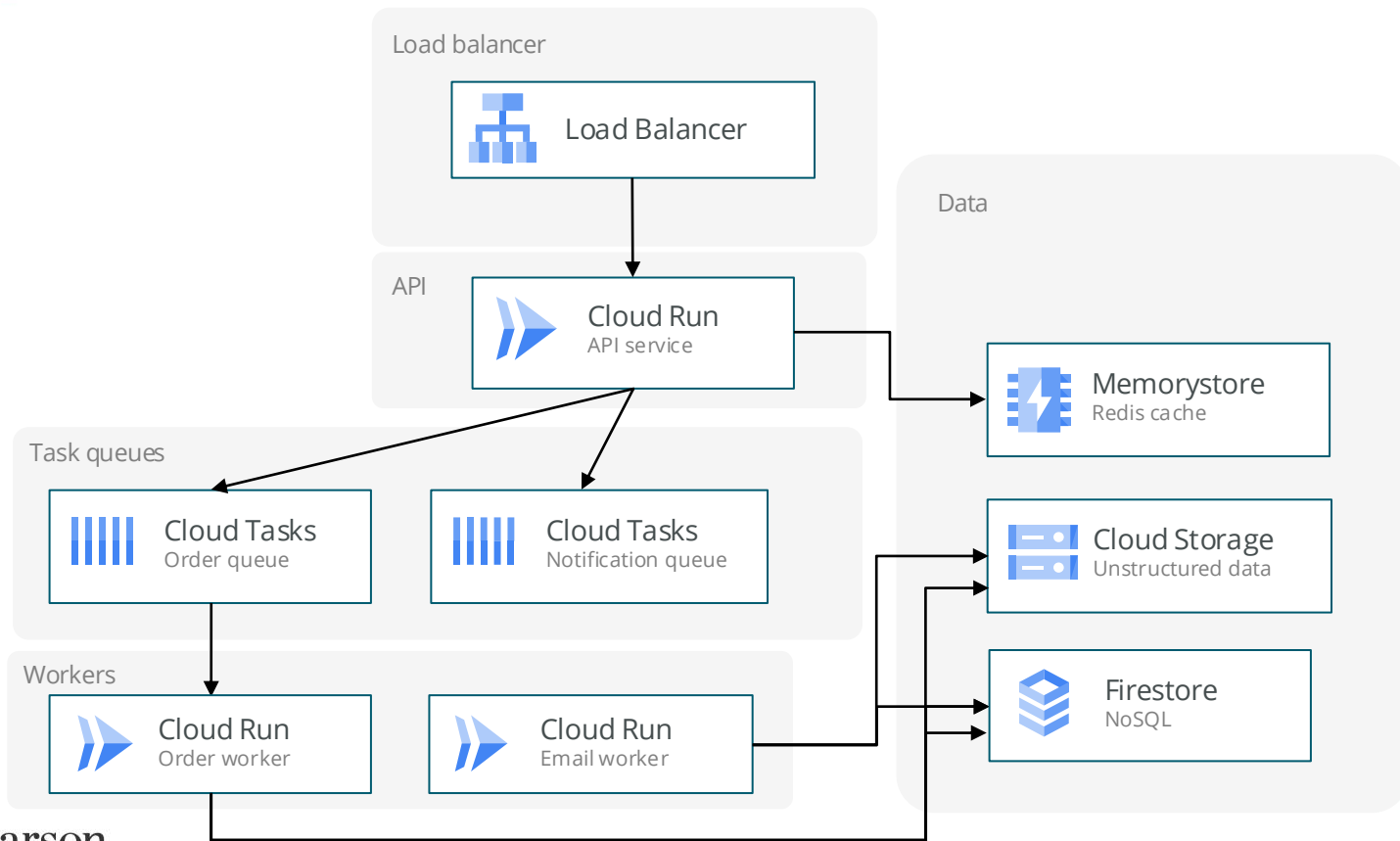
Data

Spanner, Bigtable: horizontally scalable databases

Scalable Serverless Architecture Example



Scalable Serverless Architecture Example 2



Performance Optimization

Network Latency Reduction

- Deploy close to users (regional selection)
- Use Premium Network Tier (Google backbone)
- Implement Cloud CDN for static content
- Use Global Load Balancer for anycast IPs

Application Performance

- Use connection pooling for databases
- Compress responses (gzip, Brotli)
- Optimize cold starts (Cloud Run min instances)
- Use HTTP/2 and gRPC where possible

Monitoring and Profiling

- Cloud Trace for distributed tracing
- Cloud Profiler for CPU/memory analysis
- Set latency-based SLIs and SLOs
- Use percentile metrics (p50, p95, p99)

Database Performance

- Use read replicas for read-heavy workloads
- Implement query result caching
- Optimize indexes and query plans

Premium vs Standard Tier

Premium routes via Google's backbone (lower latency). Standard uses public internet (lower cost).

Cold Start Mitigation

Use min instances in Cloud Run, provisioned concurrency in Functions, or keep-alive requests.

Gemini Cloud Assist

Gemini Cloud Assist is an AI-powered assistant integrated into Google Cloud Console that helps with design, operations, troubleshooting, and optimization across your cloud environment.

How to Access

- Available in Cloud Console (Gemini icon)
- Integrated in Cloud Shell and IDE
- Context-aware to current service/page

Use Cases for Architects

- "Design a HA architecture for e-commerce"
- "Why is my Cloud SQL slow?"
- "How can I reduce egress costs?"
- "What IAM role do I need for this function?"

Cloud Assist Preview



Your administrator is sharing all prompts and responses in this project with Google. [Learn more about data sharing](#)

Gemini Cloud Assist is an AI-powered collaborator to help you get more done faster. Get answers to your questions about Google Cloud products and best practices, and retrieve information about your cloud resources.

Learn more about how to configure and use Gemini Cloud Assist, including granting access to information about your project and resources, in the [Cloud Assist users guide](#).

What is the difference between SSD and HDD?

Help me build a GenAI app

Create a custom organization policy constraint that only allows n2d machine types

Enter a prompt



Questions Breakdown

Your company has a web-based application hosted in a single data center. As the customer base grows, customers from distant locations often complain that the website is slow. Your company has decided to move the application to Google Cloud to benefit from its global footprint.

How should you design the solution to reduce latency for customers?

A. Deploy the application to a set of virtual machines and use DNS-based load balancing

B. Use zonal managed instance groups in different zones with a regional load balancer

C. Use regional managed instance groups in different regions with a global load balancer

D. Deploy the application to a regional App Engine instance with a global load balancer

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Questions Breakdown

A financial services company has an API that processes credit card transactions. The current p99 latency is 800ms, but they need to reduce it to under 100ms to meet their SLA. The API makes calls to Cloud SQL, and profiling shows that database queries account for 70% of the latency. The queries are primarily lookups of customer and merchant data that rarely changes.

Which solution would most effectively reduce the API latency?

- A.** Migrate from Cloud SQL to Cloud Spanner for lower read latency
- B.** Add Cloud SQL read replicas and distribute queries across them
- C.** Upgrade the Cloud SQL instance to a larger machine type with more memory
- D.** Implement Memorystore for Redis to cache customer and merchant data

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Questions Breakdown

Your company is running a stateless web application on two Compute Engine instances in two availability zones. The application receives a lot of traffic during business hours and little traffic otherwise. During peak hours, several users are complaining that the application is slow and sometimes crashing. You need to redesign the solution to improve performance.

What should you do?

- A.** Deploy the application to two more instances in a separate Google Cloud region
- B.** Deploy the application to an extra instance in a new availability zone. Configure startup and shutdown scripts so that the extra instance only runs during business hours
- C.** Set up a Cloud Monitoring alert that triggers a Cloud Function to create a new instance if the average CPU utilization is high
- D.** Create an instance template and deploy the application to a managed instance group with an autoscaling policy

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Questions Breakdown

Your batch processing application runs nightly on a Compute Engine Managed Instance Group (MIG) configured with autoscaling. During periods of heavy load, you observe that the MIG fails to scale out to the desired number of instances. Cloud Logging shows QUOTA_EXCEEDED errors related to CPU resources in the region

What proactive steps should you take to ensure the application can scale reliably during peak processing times?

- A. Reduce the maxReplicas setting in the MIG's autoscaler configuration to stay below the quota
- B. Rewrite the application to use fewer CPU resources per instance
- C. Monitor your regional CPU usage and request a quota increase from Google Cloud support in advance of anticipated needs
- D. Switch the MIG to use Spot VMs, as they have separate quota limits

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Segment 4.1: Designing network, storage, and compute



Objectives

- Integrations with on-premises
- Multicloud environments
- Cloud-native networking (VPC, peering, private service connect, etc.)
- Choosing appropriate storage types

Hybrid Networking Services: VPN

HA VPN (Recommended)

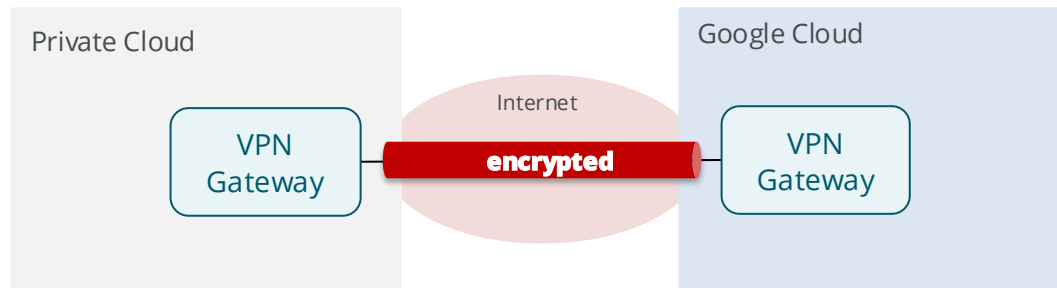
- 99.99% SLA when properly configured
- Two interfaces with two external IPs
- Requires BGP for dynamic routing
- Max 3 Gbps per tunnel

99.99% SLA Requirements

Must have tunnel from each HA VPN gateway interface to peer gateway.

Classic VPN (Legacy, partially deprecated)

- 99.9% SLA only
- Single interface, single external IP
- Use only for legacy workloads



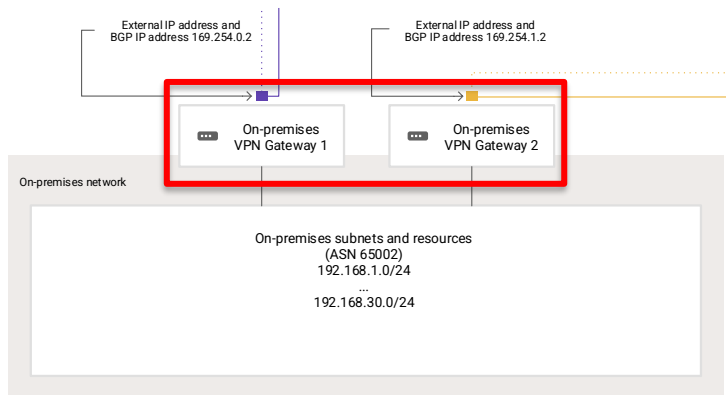
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Hybrid Networking Services: Interconnect

Dedicated Interconnect

- Direct physical connection at colocation
- 10 Gbps or 100 Gbps circuits
- Traffic never touches public internet
- Requires presence at Google PoP

Partner Interconnect

- Connect through service provider
- 50 Mbps to 50 Gbps attachments
- No colocation facility required

Data Transfer Pricing

Discounted egress: ~\$0.02/GiB vs standard internet rates.

Cross-Cloud Interconnect

- Private connectivity to AWS, Azure, Oracle
- Uses Google backbone, MACsec available

HA VPN over Interconnect

- Combines IPsec encryption + high bandwidth
- Meets compliance for encryption in transit

Network Connectivity Center

Mesh Topology

All spokes with full connectivity between VPCs

Spoke Types

- **VPC:** Connect VPC networks
- **Hybrid:** On-prem via VPN/Interconnect
- **Gateway:** Security functions (SSE)

Star Topology

Hub-and-spoke prevents direct inter-VPC traffic

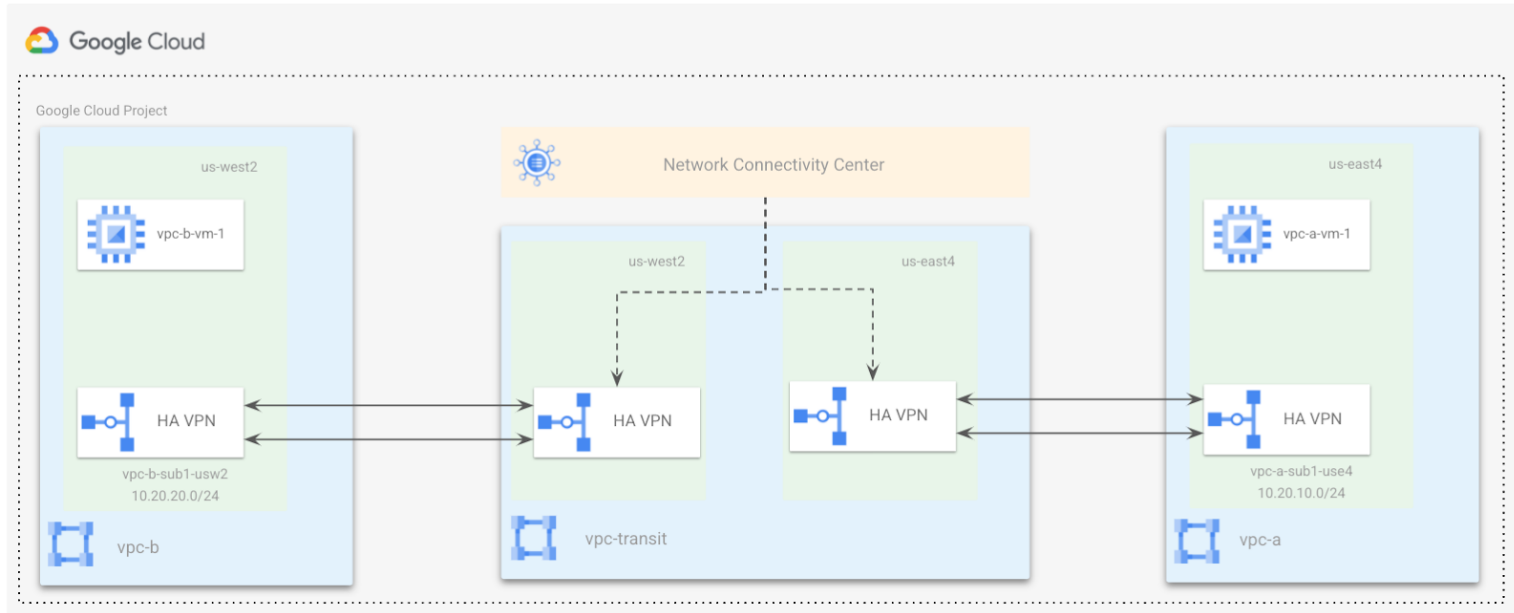
Key Benefits

- Full bandwidth between workload VPCs
- Transitive routing between all VPCs
- Cross-organization spoke support

Critical: Cloud Router cannot re-advertise routes between BGP peers. Use NCC for hub-and-spoke topologies.

Network Connectivity Center

Example hub-and-spoke architecture



Multicloud Environments

Distributed Architecture

Run applications in the computing environment that suits them best.

Google Cloud Multicloud Services

- **GKE Enterprise:** Managed K8s across clouds
- **BigQuery Omni:** Query S3/Azure in place
- **Cross-Cloud Interconnect:** Private connectivity

Redundant Architecture

Deploy same apps across multiple environments for resiliency.

Best Practices

- Use Interconnect for high data transfer volumes
- Establish common identity across environments
- Consider SLAs, performance, security, cost

GKE Enterprise (fka Anthos)

Config Management

GitOps policy

Service Mesh

Security, visibility

Connect Gateway

Secure access

Migrate

VMs to containers

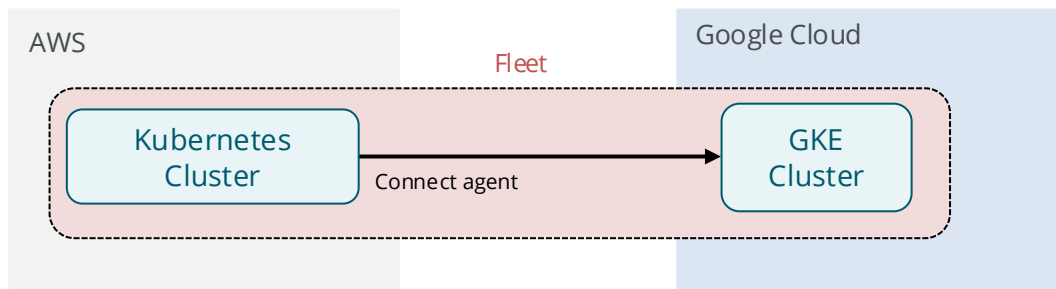
Key Capabilities

- Full Kubernetes API support everywhere
- 4-way autoscaling (cluster, node, pod, VPA)
- Multi-cluster management from single console

Supported Environments

- **GKE on Google Cloud:** Fully managed
- **GKE on AWS/Azure:** Cloud integration
- **GKE on Bare Metal/VMware**

Exam Tip: GKE Enterprise provides consistent K8s experience across clouds for unified management.



BigQuery Omni

Query data directly in AWS S3 or Azure Blob using standard BigQuery SQL. Compute spins up in target cloud. Data never moves.

Key Capabilities

- Query AWS S3 and Azure Blob in place
- Single BigQuery UI for all sources
- Cross-cloud joins with standard BQ regions

Security

- AWS IAM roles for S3 access
- Azure AD principals for Blob access

Supported Regions

- **AWS:** us-east-1, eu-west-1, ap-southeast-1
- **Azure:** eastus2, westeurope

Use Cases

- Analyze data in other clouds without moving
- Unified analytics, avoid egress costs

No Data Movement

Query in place

Single Interface

Same BigQuery SQL

Cross-Cloud Joins

Combine data

Cloud-native Networking: VPC

Auto Mode VPC

Automatic subnet per region with predefined IPs.
Good for testing.

- One subnet per region
- Fixed /20 Regional IP allocation
- Expandable up to /16

Custom Mode VPC

Full control over regions and CIDRs.
Best for production.

- No default subnets
- Full control of IP ranges
- Expandable up to any RFC 1918 sizes

Best Practice: VPCs are global, logically isolated resources. VPCs do NOT have an associated IP address range.

Cloud-native Networking: VPC

Subnets

- **Regional scope:** Spans availability zones within a region
- **Address range:** Each subnet is associated with an IP address range
- **No overlaps:** Subnets' address ranges cannot overlap with each other

Virtual, Distributed Router

- Handles internal subnet routes automatically
- You can add custom static routes manually

Virtual, Distributed Firewall

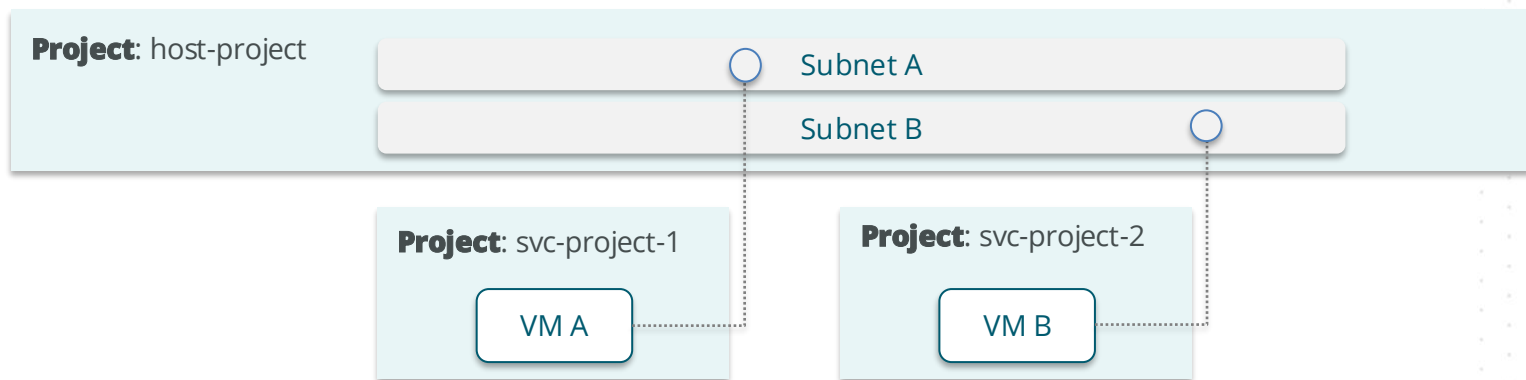
- Rules applied to every incoming and outgoing connections
- Granular control (IP, tag, service account)
- Implicit deny inbound

Hierarchical Firewall Policies

- Org/folder-level enforcement
- Cannot be overridden by lower levels

Cloud-native Networking: Shared VPC

- You can share a VPC network from one project (**host** project) to other projects (**service** projects)
- Benefits:
 - Separation of concerns
 - Enforce consistent security policies for multiple (service) projects
 - Help separate budgeting and internal cost allocation

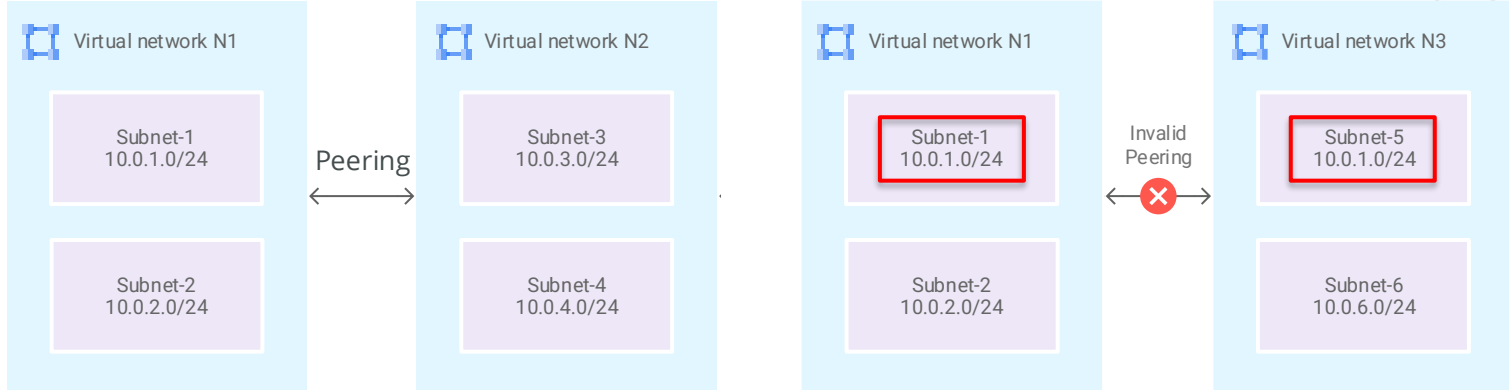


Cloud-Native Networking: VPC Network Peering

VPC Peering

- Internal IP address connectivity across VPCs
- Can span different organizations
- Non-transitive (A-B, A-C $\neq$ B-C)
- Both sides must configure peering

Note: Networks with overlapping ranges cannot be peered



Private Access Options

Private Google Access (PGA)

Access Google APIs from VMs with internal IPs only.

- Configured per subnet
- Access Cloud Storage, BigQuery, etc.

Private Service Access (PSA)

Private IPs for Google-managed services.

- Uses VPC peering internally
- For Cloud SQL, Memorystore, Vertex AI

Private Service Connect (PSC)

Private, secure access to services and SaaS.

- Creates endpoint in your VPC
- Connect to Google or third-party

When to Use Each

- **PGA:** VMs with internal IPs need Google APIs (Storage, BigQuery)
- **PSA:** Need private IPs for managed services (Cloud SQL, Redis)
- **PSC:** Fine-grained control, third-party SaaS, publish your services

Private Google Access

Private Google Access enables VMs with **only internal IP addresses** to reach Google APIs and services. Traffic stays on Google's network and never traverses the public internet.

How It Works

- Enabled per subnet (not per VM or VPC)
- VMs send requests to Google API public IPs
- Google routes traffic internally (no internet egress)
- Responses return via same internal path

Supported Services

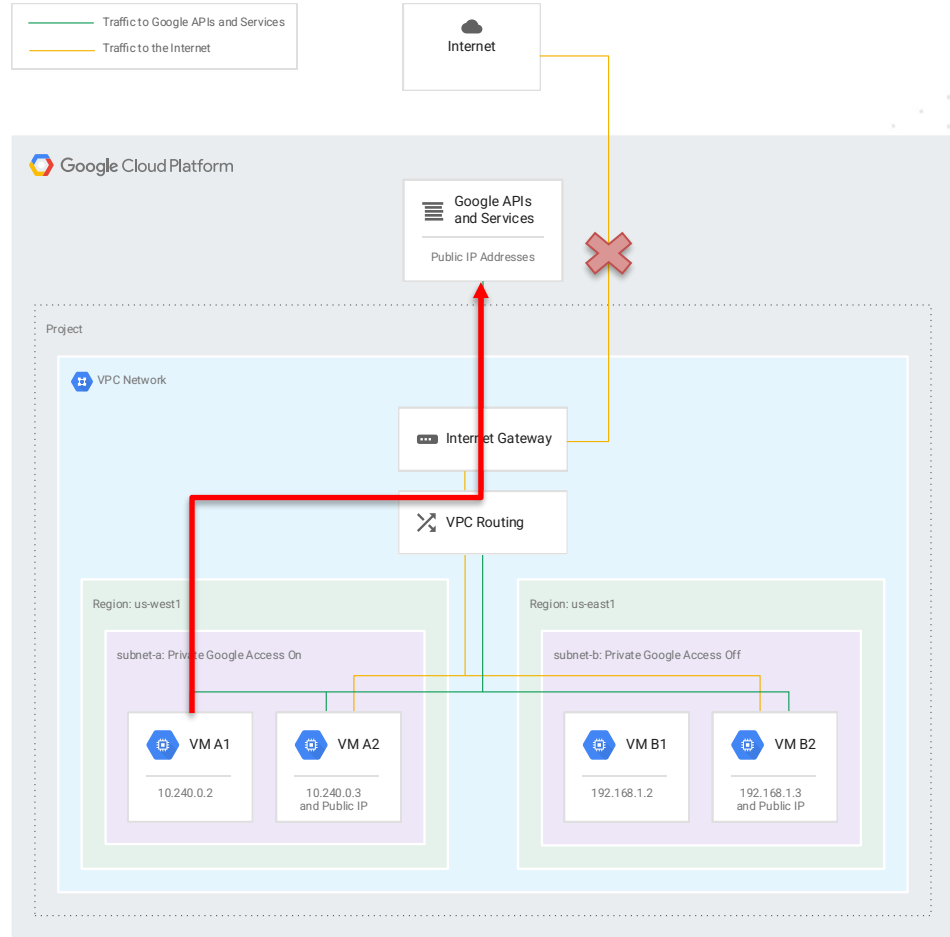
- Cloud Storage, BigQuery, Pub/Sub
- Container Registry, Artifact Registry
- Cloud Functions, Cloud Run
- Most Google APIs (maps.googleapis.com, etc.)

Exam Tip: PGA is enabled per-subnet.

Cloud-native Networking: Private Google Access

Important: Private Google Access has no effect on instances that have external IP addresses

VM A1 is the only one
that benefits from PGA



Private Service Connect

Private Service Connect creates **private endpoints** in your VPC to access Google services, third-party SaaS, or publish your own services. Unlike VPC peering, PSC uses forwarding rules with private IPs you control.

Access Google APIs

Single endpoint IP for all Google APIs

Access Managed Services

Cloud SQL, AlloyDB, Memorystore

Third-Party SaaS

MongoDB, Confluent, Snowflake

Publish Your Services

Expose to other VPCs/orgs

Key Components

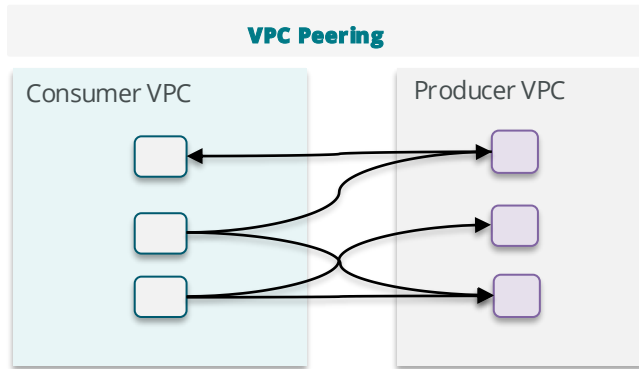
- **Endpoint:** Forwarding rule with private IP in your VPC
- **Service Attachment:** Producer-side resource exposing the service
- **Connection:** Links endpoint to service attachment
- **NAT Subnet:** IP range for producer-side connections

PSC for Google APIs

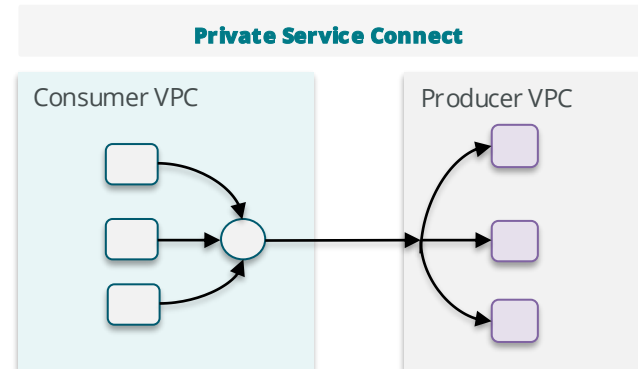
- Create endpoint: global or regional
- Single IP accesses all Google APIs
- Configure DNS: *.googleapis.com → endpoint IP
- Automatic with managed DNS zones

VPC Peering vs Private Service Connect (PSC)

Exam Tip: PSC is the modern approach for private connectivity. Choose PSC over VPC peering when you need IP address control, cross-org access, or want to avoid peering limits.



Network-to-network, many-to-many communication



Service-oriented model, uni-directional

PSC Benefits Over VPC Peering

- No IP overlap concerns (you choose endpoint IP)
- No transitive routing issues
- Fine-grained access control per endpoint
- Works across organizations

Cloud-native Networking: Private Service Connect



Consumer VPC

us-central-1

subnet-101
10.10.1.0/24

subnet-102
10.100.7.0/24



VM



VM



VM



VM

Private Service
Connect endpoint
(forwarding rule)

IP: 10.100.7.9
Target: service-1



Producer VPC

us-central-1

subnet-202
10.101.0.0/16

Service attachment

Name: service-1
Subnet:
192.168.0.0/16

Forwarding rule
IP: 10.101.0.7

Backend service

Health check

Instance group



Instance



Instance



Instance

Cloud-native Networking: VPC Service Controls

VPC Service Controls create **security perimeters** around Google Cloud resources to prevent data exfiltration. Even users with IAM permissions cannot move data outside the perimeter.

Service Perimeter

Boundary around projects and services

Access Levels

Conditions for allowing access

Ingress/Egress Rules

Controlled exceptions to perimeter

What VPC-SC Protects Against

- Data exfiltration via compromised credentials
- Copying data to unauthorized projects
- Accessing resources from outside perimeter
- Insider threats with valid IAM permissions

Access Levels (Context-Aware)

- IP address ranges (corporate network)
- Device security posture (managed devices)
- User identity and group membership
- Geographic region restrictions

Exam Tip: VPC-SC prevents data exfiltration by going beyond IAM permissions and looking at more granular access context.

Cloud NAT

Cloud NAT

- Outbound internet for VMs without external IPs
- Proxy-less design at SDN layer (no bottleneck)
- 64,512 source ports per external IP
- Monitor nat_allocation_failed metric

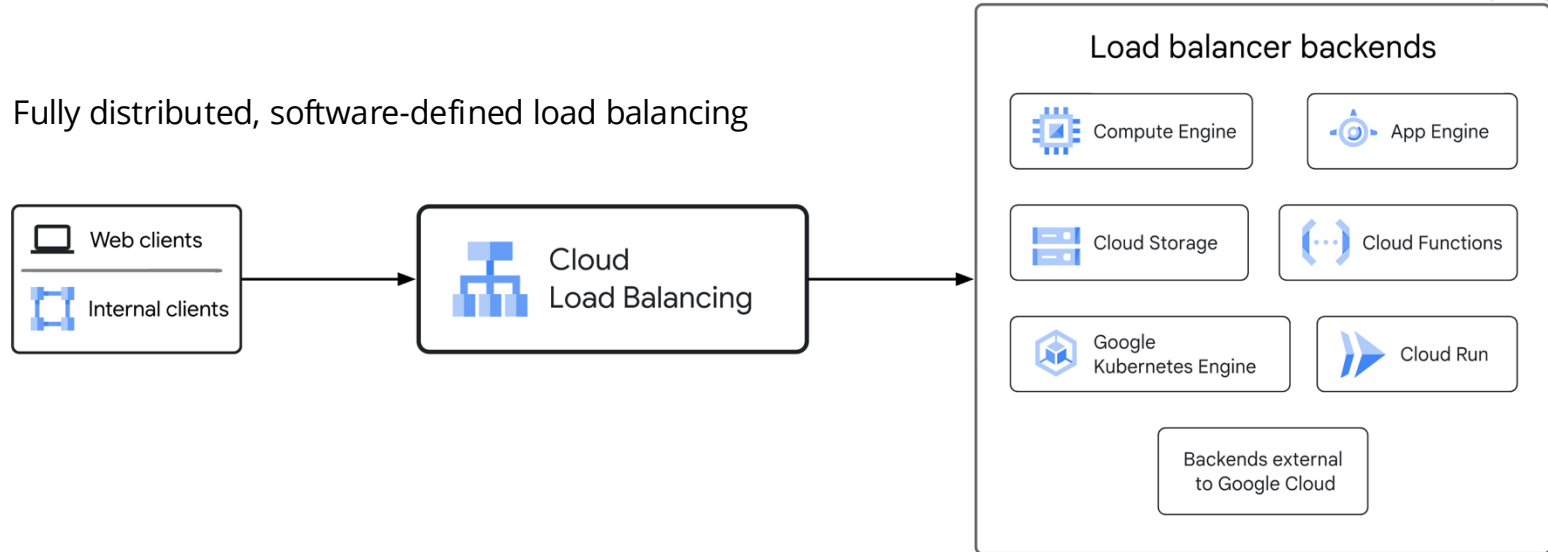


Cloud NAT Best Practices

- Use auto-allocation for standard use cases
- Static IPs for allowlisting/compliance

Cloud-native Networking: Load Balancing

Fully distributed, software-defined load balancing



Cloud-native Networking: Load Balancing

 Cloud Load Balancing	External (Accepts internet traffic)	Internal (Accepts only RFC 1918 traffic)
Application Load Balancers HTTPS Layer 7 load balancing	• global external • regional external • classic	• cross-region internal • regional internal
Network Load Balancers TCP/SSL/Other Layer 4 load balancing	Proxy Network Load Balancers	
	• global external • regional external • classic	• cross-region internal • regional internal
	Passthrough Network Load Balancers	
	• regional external	• regional internal

Just remember this

- Public web app regional or global: Regional/Global External HTTP/S LB
- Non-web apps: Regional/Global Internal/External Proxy Network LB
- Preserve client IP or UDP traffic: Passthrough Network LB

Managed Database Services

SQL Databases



Cloud SQL
MySQL, Postgres, SQL Server



Spanner
Postgres, Google SQL



AlloyDB
Postgres



BigQuery
Google SQL (warehouse)

NoSQL Databases



Firestore
Document-style



Memorystore
Redis, Memcached, Valkey



Bigtable
Key-value, wide-column, H-base compatible

Choosing Database Services

Scale Considerations

- **< 10 TB:** Cloud SQL, AlloyDB, Firestore
- **10 TB - PB:** Spanner, Bigtable, BigQuery
- **Global distribution:** Spanner or Firestore

Cloud SQL vs AlloyDB

- Both PostgreSQL-compatible
- Cloud SQL: Supports MySQL + SQL Server too
- AlloyDB: 4x transactions, built-in analytics

Consistency Model

- **Strong:** Spanner, Firestore, AlloyDB, Cloud SQL
- **Eventual:** Bigtable (prioritizes throughput)

Caching Layer

- **Memorystore:** Redis/Memcached in-memory
- Sub-millisecond access for hot data
- Complements primary databases

Exam Tip: Match requirements to the simplest service. Cloud SQL first, AlloyDB for PostgreSQL perf, Spanner only when global consistency is needed.

Database Storage Design

Cloud SQL Storage

- SSD (default) or HDD storage
- Auto-increase enabled (cannot decrease)

Spanner Storage

- Server and storage resources determined by processing units / nodes
- Tiered storage: HDD 80% cheaper

AlloyDB Storage

- Disaggregated compute and storage
- 4x transactions, 100x analytics

Bigtable Design

- Row key design is critical (avoid sequential keys to prevent hotspots)
- SSD or HDD storage

Selection Guide

- Lift-and-shift MySQL/PostgreSQL: **Cloud SQL**
- High-perf PostgreSQL: **AlloyDB** | Global 99.999%: **Spanner**

Cloud Storage

Cloud Storage is Google Cloud's object storage service. Objects are stored in **buckets** with a flat namespace (folders are simulated via prefixes). Strongly consistent for all operations.

Access Control

- **Uniform:** IAM only (recommended)
- **Fine-grained:** IAM + per-object ACLs
- Signed URLs for temporary public access
- VPC Service Controls for exfiltration prevention

Common Use Cases

- Data lake / analytics staging (BigQuery, Dataflow)
- Backup, disaster recovery, archival
- Static website hosting, ML model artifacts

Performance Considerations

- 5,000 writes/sec and 8,000 reads/sec per prefix
- Use parallel composite uploads for large objects
- Avoid sequential naming (distribute key prefixes)
- Turbo replication: RPO of 15 min for dual-region

Data Transfer Options

- **gsutil/gcloud:** CLI for ad-hoc transfers
- **Storage Transfer Service:** Scheduled, cross-cloud
- **Transfer Appliance:** Offline, petabyte-scale

Exam Tip: Bucket location cannot be changed after creation. Choose region to co-locate with compute; dual-region for HA with controlled geo-placement.

Block Storage Options

Zonal vs Regional PD

- **Zonal:** Single zone, higher perf, lower cost
- **Regional:** Sync replication, survives zone failure

Performance Scaling

- IOPS/throughput scales with disk size
- Also limited by VM vCPU count

Hyperdisk Advantages

- Independent scaling: capacity, IOPS, throughput
- Hyperdisk ML: Multi-attach up to 100 VMs

Snapshots

- Incremental (only changed blocks)
- Global resource (cross-region)

Local SSD

Highest performance (2.4M read IOPS). Ephemeral - data lost on VM stop. Use for caches only.

File Storage Options

Filestore Features

- NFS v3 protocol (native Linux/Windows)
- GKE CSI driver integration
- Performance scales with capacity

When to Use Filestore

- POSIX compliance required
- Low latency file access needed
- Concurrent file locking required

Cloud Storage FUSE

- Mount Cloud Storage as file system
- 9x faster model load (parallel downloads)

When to Use Cloud Storage

- Unlimited scale needed
- Cost optimization with lifecycle

Exam Tip: Filestore for NFS/POSIX. Cloud Storage for object storage economics.

Segment 4.2: Designing network, storage, and compute



Objectives

- Choosing data processing technologies
- Overview of Google Cloud AI and machine learning solutions
- Choosing compute platforms and resources
- Sample exam questions

Data Processing Services

Dataflow (Apache Beam)

- Fully managed, serverless processing
- Unified batch and streaming pipelines
- Auto-scaling up to 4K workers per job
- Advanced windowing and watermarks
- No cluster management required
- Best for new pipelines and real-time ETL

Dataproc (Hadoop/Spark)

- Managed Hadoop, Spark, Flink, Presto
- 90-second cluster start/scale/shutdown
- Supports 30+ open-source tools
- Full control over cluster configuration
- Spot VMs for cost savings
- Best for migrating existing Spark/Hadoop jobs

When to Choose

- **Dataflow:** New pipelines, streaming, serverless, Apache Beam, no ops team
- **Dataproc:** Existing Spark/Hadoop jobs, need cluster control, lift-and-shift migration

Messaging and Event Streaming

Pub/Sub is a fully managed, real-time messaging service for asynchronous communication. Messages are replicated across 3 zones. Global routing delivers messages from any region.

Fan-Out

One topic, many subscribers

Load Balancing

Multiple workers, one subscription

Event-Driven

Decouple microservices

Key Features

- At-least-once or exactly-once delivery
- Push and pull subscription models
- Message ordering via ordering keys
- Dead-letter topics for failed messages
- Server-side message filtering
- Retention up to 31 days (default 7 days)

Common Use Cases

- Streaming source for Dataflow pipelines
- IoT data ingestion from devices
- Event-driven Cloud Functions/Run triggers
- Log aggregation and monitoring
- Cross-region async communication

Pub/Sub Lite

- Deprecated: discontinuing March 2026
- Migrate to standard Pub/Sub or Kafka

Exam Tip: Pub/Sub + Dataflow is the standard pattern for real-time streaming analytics.

Push vs Pull Mode

Cloud Pub/Sub: Push mode

- Pub/Sub server initiates request to deliver messages
- One response is returned in each request
- Automatic flow control based on acknowledgement rate



Use cases: multiple topics, one webhook, Cloud Run Functions subscribers

Cloud Pub/Sub: Pull mode

- Subscriber client initiates requests to retrieve messages
- Can have multiple responses returned
- The subscriber client controls the rate of delivery (flow control)



Use cases: large volume of messages, throughput is critical

Managed Services for Apache Kafka

Fully managed Apache Kafka (3.7, KRaft mode) running across 3 zones for HA. Uses unmodified open-source Kafka, so existing applications work without changes. Automatic security patching and rolling updates.

Architecture & Scaling

- Specify total vCPU + RAM; brokers auto-provisioned
- Vertical scaling up to 15 vCPU/broker, then horizontal
- Tiered storage: SSD + Cloud Storage for segments

When to Choose over Pub/Sub

- Existing Kafka workloads (lift-and-shift)
- Need Kafka protocol compatibility
- Require consumer groups and partition control
- Complex stream processing with Kafka Streams

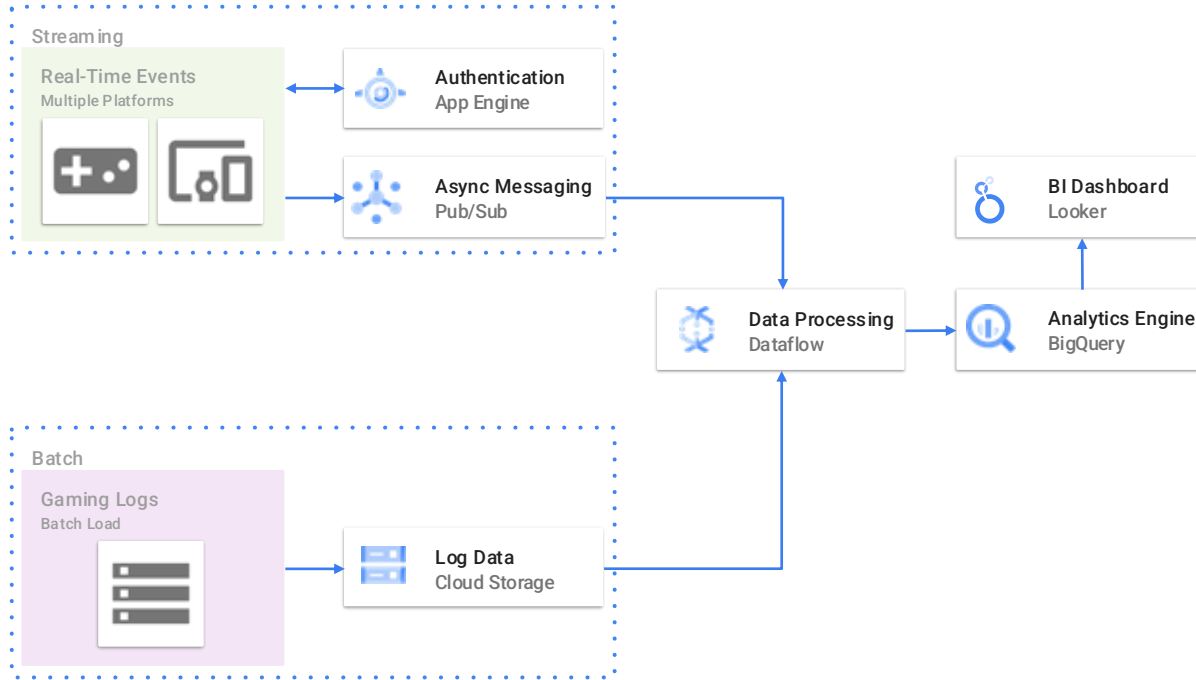
Integrations

- Kafka Connect: BigQuery, Cloud Storage, Pub/Sub
- Confluent Schema Registry (Avro, Protobuf)
- Cloud Monitoring metrics, Cloud Logging
- Terraform, gcloud CLI, REST/gRPC APIs

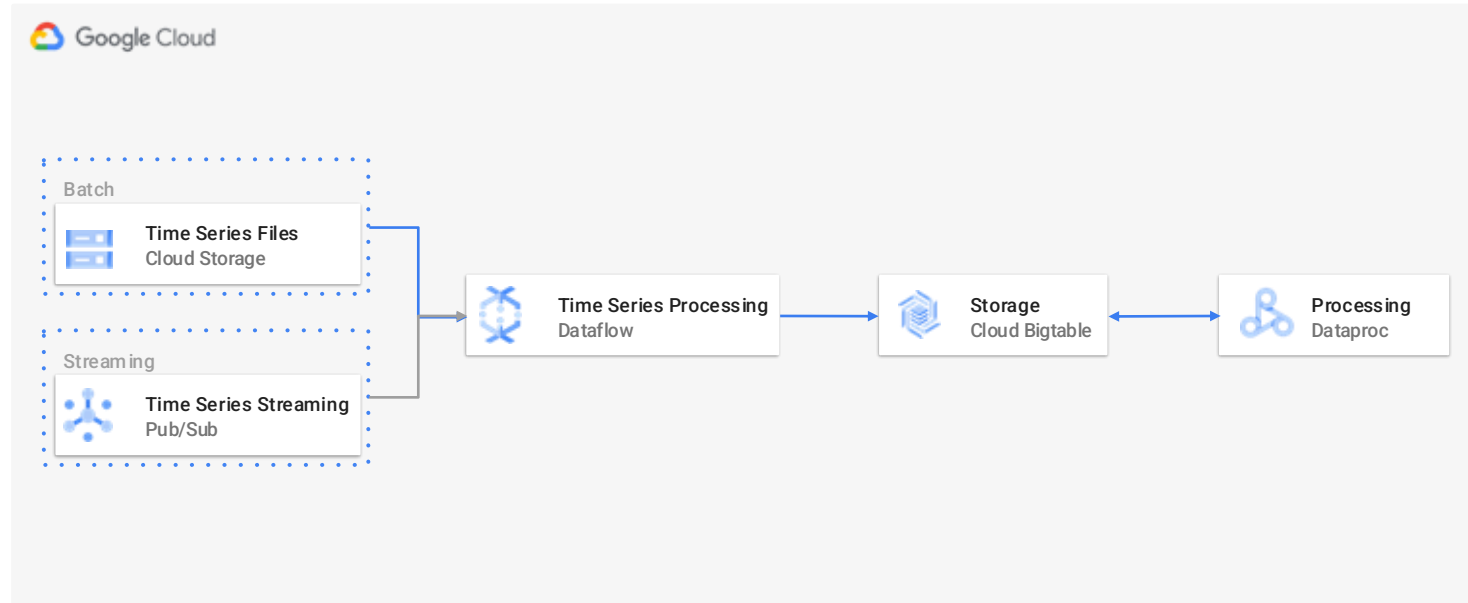


Exam Tip: Choose Managed Kafka for existing Kafka workloads needing lift-and-shift. For new event-driven architectures, prefer Pub/Sub (serverless, no cluster mgmt).

Streaming and Batch Data Pipeline Example



Bid Data Processing: Time Series Analysis



Other Data Processing Services

Cloud Dataprep

- Integrated partner service operated by Trifacta
- Visual data wrangling tool to explore, combine, and transform data
- A set of transformation steps is called a “recipe”

Cloud Data Fusion

- Powered by the open source project CDAP (with pipeline portability)
- Visual data wrangling and data integration/pipelining tool (code-free ETL)

Google Cloud AI/ML Landscape

Google Cloud provides AI/ML services at every level of abstraction, from pre-trained APIs requiring zero ML expertise to fully custom training on TPU pods. The unified platform is **Vertex AI**.

Pre-Trained APIs (No ML Expertise Required)



Translate API



Speech-to-Text



Natural Language



Vision

AutoML (Your Data, Automated Training)

No-code/low-code model training via Vertex AI. Image, tabular, text, video. Automates feature engineering, architecture search, and hyperparameter tuning.

Custom Training (Full Control)

TensorFlow, PyTorch, JAX, scikit-learn, XGBoost. Serverless or managed clusters on GPUs/TPUs. Vertex AI Pipelines for MLOps.

Generative AI

Gemini model family, Model Garden (200+ models), Vertex AI Studio for prompt engineering, Agent Builder for AI agents.

BigQuery ML

Train and serve ML models using SQL directly in BigQuery. Ideal for SQL-proficient teams with data already in BigQuery.

Exam Tip: Know Vertex AI as the unified platform and when to use each abstraction level.

Vertex AI: Unified AI/ML Platform

Vertex AI is Google Cloud's fully managed, unified AI development platform. It brings together data engineering, data science, and ML engineering workflows

Core Components

- **Model Garden:** 200+ models (Gemini, Claude, Llama, Gemma)
- **Vertex AI Studio:** Prompt, test, and tune foundation models
- **Agent Builder:** Low-code visual designer for AI agents
- **Colab Enterprise:** Managed notebook environment
- **TensorBoard:** Experiment tracking and visualization

Training Options

- **AutoML:** No-code training on your labeled data
- **Custom Training:** Full control via TF, PyTorch, JAX
- **Serverless:** On-demand, fully managed environment
- **Training Clusters:** Reserved accelerator capacity

Serving & Deployment

- Online prediction: low-latency endpoints
- Batch prediction: async, 50% cheaper
- Dedicated endpoints: streaming, gRPC, OpenAI-compatible
- Model Registry for versioning and lineage

MLOps

- **Pipelines:** ML workflow as DAG (KFP or TFX)
- **Feature Store:** Centralized feature management
- **Experiments:** Track hyperparameters and metrics
- **Explainable AI:** Model interpretability

Exam Tip: Vertex AI should be your default answer for ML platform questions.

Industry Solutions

Document AI

- Pre-trained parsers for invoices, receipts, purchase orders, and shipping documents
- Custom Document Extractor for proprietary forms (returns, claims)
- OCR + entity extraction in a single API call
- Warehouse Document AI for logistics/shipping labels
- Integrates with Cloud Storage triggers and Workflows for automated pipelines
- Human-in-the-loop review via Document AI Workbench for low-confidence extractions

Vertex AI Search for commerce

- Managed semantic/hybrid search over product catalogs, support docs, policies
- Supports structured (product DB), unstructured (PDFs, HTML), and website data stores
- Built-in ranking, spell correction, faceting, and autocomplete
- Recommendations AI integration for personalized product suggestions
- Media recommendations for content-heavy storefronts (video, articles)
- Follow-up search with multi-turn conversations for customer self-service

Generative AI on Vertex AI

Vertex AI provides access to Google's Gemini model family and 200+ third-party/open models via Model Garden. Vertex AI Studio enables prompt engineering, testing, and model tuning.

Third-Party Models

- Anthropic Claude family
- Meta Llama 3.2
- Google Gemma (open weights)
- DeepSeek, Mistral, and more

Vertex AI Studio

- Prompt and test with Gemini using text, code, images
- Adapter tuning, RLHF for model customization
- Style/subject tuning for image generation
- Built-in prompt sharing without admin setup

Agent Builder

- Low-code visual designer for AI agents
- Agent Engine Sessions and Memory Bank (GA)
- Grounding with Google Search or enterprise data
- Integrate with existing APIs and tools

Exam Tip: Know that Model Garden provides both first-party (Gemini) and third-party models. Vertex AI Studio is for prompt engineering; Agent Builder is for building AI agents.

BigQuery ML

BigQuery ML lets analysts build, train, and deploy ML models directly in BigQuery using standard SQL. No need for separate tools, data movement, or ML frameworks — the data stays where it is.

Supported Model Types

- Linear and logistic regression
- Deep neural networks (DNN)
- XGBoost (boosted trees)
- Time-series forecasting (ARIMA+)
- K-means clustering
- Matrix factorization, PCA
- Imported TensorFlow/ONNX models
- Remote models on Vertex AI

Key Features

- **BQML Inference Engine:** Run inferences directly in BQ
- **Object Tables:** Analyze unstructured data from GCS
- **Hugging Face:** 180K+ models via SQL
- **ML.TRANSFORM:** Feature preprocessing
- Console-based model creation (Preview)

When to Use BQML vs Vertex AI

- **BQML:** Data in BQ, SQL team, standard models, fast prototyping
- **Vertex AI:** Custom architectures, GPU/TPU, advanced MLOps

Exam Tip: BQML is the answer when "SQL analyst" or "no ML expertise" appears with data already in BigQuery. For advanced ML, use Vertex AI.

Choosing the Right ML Solution

Decision Flow

- Common task (OCR, sentiment, translation)? → **Pre-trained API**
- Data in BigQuery + SQL team? → **BigQuery ML**
- Need domain-specific model, limited ML team? → **AutoML**
- Fine-grained architecture control needed? → **Custom Training**
- Need generative AI (text, images, agents)? → **Gemini / Model Garden**

Exam Tip: "No ML expertise" → Pre-trained API or AutoML. "TPU" mentioned → Custom training. Everything is unified under Vertex AI. Default to managed services over self-managed.

Compute Engine

General-Purpose Series

- **E2:** Cost-optimized, shared-core, up to 32 vCPUs. Dev/test, microservices
- **N2:** Intel, up to 128 vCPUs, 8 GB/vCPU. Custom types, SUDs eligible
- **N2D:** AMD EPYC, up to 224 vCPUs. Custom types, SUDs eligible
- **N4:** Intel Emerald Rapids + Titanium. DDR5, Hyperdisk only
- **N4A:** Google Axion (Arm). Best price-performance for scale-out
- **C4:** Intel Emerald Rapids + Titanium. Up to 200 Gbps Tier_1

Custom Machine Types

- Available on N1, N2, N2D, N4 series
- 1 to 96 vCPUs, up to 8 GB/vCPU
- Extended memory option beyond 8 GB/vCPU
- Key cost optimization — right-size instead of over-provisioning

Compute-Optimized Series

- **C2:** Intel Cascade Lake, 3.8 GHz turbo. Gaming, HPC. SUDs eligible
- **C2D:** AMD EPYC Milan, up to 112 vCPUs
- **C3:** Intel Sapphire Rapids + Titanium, up to 176 vCPUs, DDR5
- **C3D:** AMD EPYC Genoa + Titanium, up to 360 vCPUs
- **H3:** Fixed 88 vCPUs, HPC only. Compact placement for MPI

Live Migration

- GCE transparently moves VMs during maintenance
- Zero downtime — default for most VMs
- Configurable: migrate (default) or terminate

Exam Tip: Live migration is a GCE differentiator — zero downtime for maintenance. Use custom machine types to avoid paying for unused resources.

Compute Engine

Memory-Optimized

- **M1:** Intel Skylake, up to 4 TB RAM. SUDs eligible
- **M2:** Intel Cascade Lake, up to 12 TB RAM. Largest memory. SUDs eligible
- **M3:** Intel Ice Lake, up to 4 TB, 128 vCPUs. NVMe + gVNIC only
- **X4:** Intel Sapphire Rapids + Titanium. Ultra-high memory

Memory-Optimized Use Cases

- SAP HANA (certified M2/M3 shapes)
- In-memory databases (Redis, Memcached)
- Real-time analytics (OLAP)
- Genomics and scientific simulations

Accelerator-Optimized (GPU)

- **A2:** NVIDIA A100. General ML training/inference
- **A3 High:** NVIDIA H100. Training and serving
- **A3 Mega:** NVIDIA H200. Large-scale training
- **A4:** NVIDIA B200 Blackwell. Foundation models
- **G2:** NVIDIA L4. Cost-optimized inference, FP8 support
- **G4:** NVIDIA RTX PRO 6000 Blackwell. Next-gen graphics

GPU Considerations

- GPUs do NOT support live migration
- Spot GPU VMs: preempted during maintenance, use MIG to recreate
- Run on GKE with GPU node pools for orchestration
- Use Kueue for multi-tenant GPU scheduling

Exam Tip: M..-> Memory, C..-> Compute, A..-> Accelerator (AI/ML, gaming) with few exceptions

Instance Groups and Autoscaling

Managed Instance Groups (MIGs) create identical VMs from an instance template. They provide autoscaling, autohealing, rolling updates, and multi-zone distribution. **Unmanaged groups** hold heterogeneous pre-existing VMs (load balancing only, no autoscaling).

MIG Features

- **Stateless MIGs:** Web servers, APIs — full autoscaling
- **Stateful MIGs:** Persistent disks/IPs — no autoscaling
- **Regional MIGs:** Spread across zones for HA
- Rolling updates with canary deployments
- Autohealing via health checks

Unmanaged Instance Groups

- Heterogeneous VMs, different configs
- No autoscaling, autohealing, or updates
- Only for load balancing pre-existing VMs
- Almost never the right exam answer

Autoscaling Signals

- CPU utilization
- HTTP load balancing capacity
- Pub/Sub queue depth
- Cloud Monitoring / custom metrics
- Multiple signals → takes largest recommendation

Advanced Autoscaling

- **Stabilization:** 10-min window, uses peak load
- **Predictive:** Scales proactively from historical data
- **Scheduled:** Scale on known traffic patterns
- **Scale-in controls:** Min instances, cooldown, max rate

Exam Tip: Regional MIGs for HA across zones is the most common exam answer. Stateful MIGs for stateful apps.

Sole-Tenant Nodes and Spot VMs

Sole-Tenant Nodes

- Dedicated physical servers for your project only
- **BYOL:** Per-core/processor license compliance
- **Compliance:** Physical isolation requirements
- Pay for full node capacity + tenancy premium
- CPU overcommit on N1, N2, N2D, N4 only
- Cannot use Spot/preemptible VMs on sole-tenant

Spot VM

- Offered at a steep discount in periods of excess capacity
- Ideal for fault-tolerant batch, CI/CD, data processing
- Use MIGs to auto-recreate preempted VMs (when available)

Exam Tip: Sole-tenant = BYOL or compliance. Spot VMs for non-mission critical workloads only (e.g. nightly batch processing)

Google Kubernetes Engine

- Managed [Kubernetes](#) platform
- High-availability control plane
- Cluster autoscaling
- Two modes of operations: **Standard** and **Autopilot**



When to Choose

- **Autopilot:** Most production workloads, low-ops teams, stateless apps, minimize operational overhead
- **Standard:** Complex stateful workloads, full control needed, custom node configs, Windows containers

Exam Tip: "Minimize operational overhead" → Autopilot. Google recommends Autopilot as default.

Serverless Compute

Cloud Run

- Fully managed platform for stateless containers
- Any language, library, or binary (OCI containers)
- Scales to zero (pay nothing when idle)
- Request timeout up to 60 minutes
- Configurable concurrency (default 80/instance)
- VPC connectors for private network access
- Min instances to eliminate cold starts

Cloud (Run) Functions

- Event-driven Functions-as-a-Service (FaaS)
- Triggers: HTTP, Pub/Sub, GCS, Firestore, Eventarc
- Scales to zero
- 2nd gen built on Cloud Run
- Timeout: 60 min (2nd gen)
- Limited languages: Node, Python, Go, Java, .NET

Exam Tip: "Containers" or "portability" → Cloud Run. "Event-driven single function" → Cloud Functions. Cloud Functions 2nd gen is built on Cloud Run.

Serverless Compute: App Engine

Standard Environment

- Pre-configured runtimes (Python, Java, Node, Go, etc.)
- Scales from zero to thousands rapidly
- Custom autoscaling algorithm
- Fastest deployments (seconds)

Flexible Environment

- Docker containers on Compute Engine VMs
- Any language, custom runtimes, SSH access
- Cannot scale to zero (min 1 instance)
- Uses Compute Engine Autoscaler

Exam Tip: Standard = rapid scale, lowest cost, limited runtimes. Flexible = custom containers on VMs, no scale-to-zero. For new apps, prefer Cloud Run over App Engine Flexible.

Mapping Compute Needs to Platform

Decision Flow

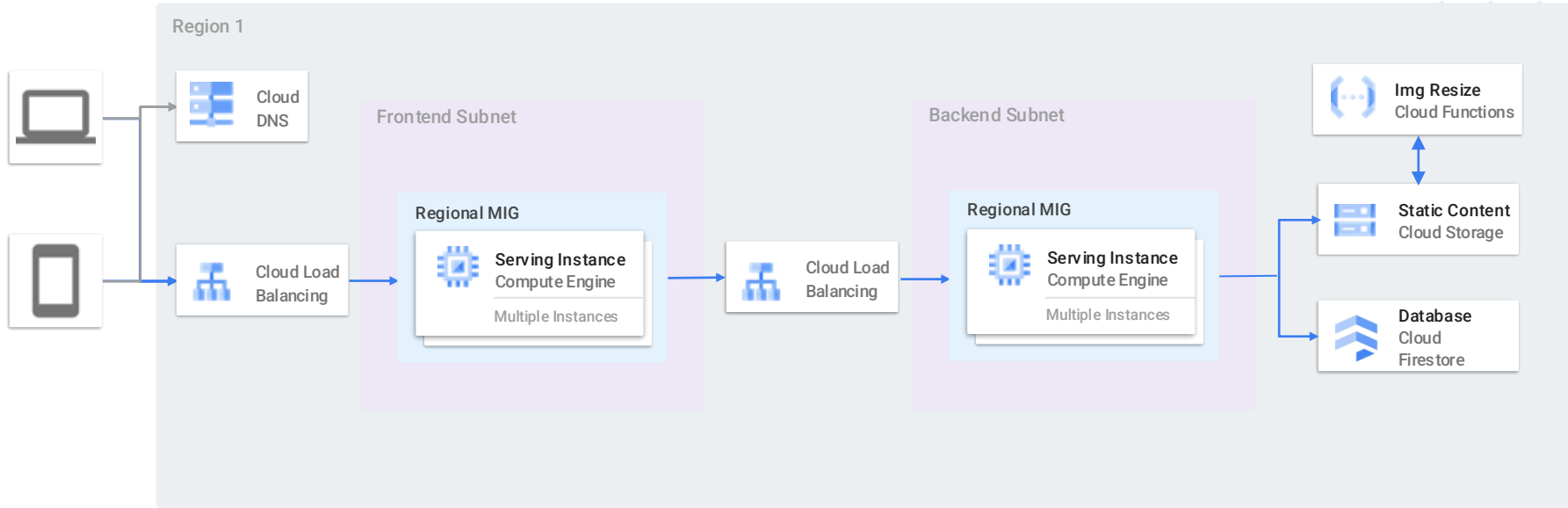
- Need full OS/hardware control? → **Compute Engine** **Keywords:** full control; access to OS
- Containerized + K8s orchestration? → **GKE (Autopilot or Standard)** **Keywords:** container, portability
"minimize ops overhead" → GKE Autopilot, Cloud Run
existing K8s, container orchestration -> GKE Standard
- Containerized but no K8s needed? → **Cloud Run** **Keywords:** event-driven, lightweight ETL, webhook
- Event-driven function logic? → **Cloud Functions**
- Web application (legacy or simple)? → **App Engine** **Keywords:** serverless app platform, common runtime

Example: Designing Network, Storage, and Compute

Requirements

- Regional 3-tier web app
- IaaS Compute with Debian Linux
- A document database
- Object storage for images
- Event-driven processing for new images (image resizing)

Example: Designing Network, Storage, and Compute



Questions Breakdown

A financial services company is migrating their data center to Google Cloud. They have strict compliance requirements mandating that traffic between on-premises and cloud never traverse the public internet. They need 50 Gbps of bandwidth and have existing relationships with several network service providers, but do not have a presence in any Google colocation facilities.

Which connectivity option should you recommend?

A. Dedicated Interconnect with 5x 10 Gbps connections

B. Partner Interconnect with 50 Gbps VLAN attachments

C. HA VPN with multiple tunnels aggregated via ECMP

D. Cloud VPN over Dedicated Interconnect

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Questions Breakdown

Your organization has 15 projects across 3 teams, each with their own VPC networks. You need to enable communication between all VPCs while maintaining centralized network administration and a single Cloud Interconnect connection to your on-premises data center.

Which networking architecture should you implement?

- A.** Create VPC peering connections between all 15 VPCs
- B.** Use Shared VPC with one host project and 14 service projects
- C.** Deploy a Cloud Interconnect connection in each VPC
- D.** Use Network Connectivity Center with VPC spokes for each project

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Questions Breakdown

A global e-commerce company has data stored in AWS S3 buckets in us-east-1 and Google Cloud Storage in us-central1. The analytics team needs to run daily reports combining data from both sources using SQL. They want to avoid data movement to minimize egress costs and maintain data freshness.

Which solution should you recommend?

- A.** Use Storage Transfer Service to copy S3 data to Cloud Storage daily, then query with BigQuery
- B.** Use Dataflow to stream data from S3 to BigQuery tables before running reports
- C.** Set up a Dataproc cluster with connectors for both S3 and Cloud Storage
- D.** Use BigQuery Omni to query S3 data in place and join with Cloud Storage data in BigQuery

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Questions Breakdown

A new workload will require a mix of batch and stream processing of data on GCP. The data are to be sourced from Cloud Storage and Cloud Pub/Sub and need to be transformed before being ingested in BigQuery for analysis.

Which technology should you choose?

A. Cloud Dataproc

B. Cloud SQL

C. Cloud Dataflow

D. BigQuery Data Transfer Service

Questions Breakdown

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C. Cloud Dataflow

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Questions Breakdown

A retail company wants to analyze customer reviews to determine sentiment and extract key topics. The team consists of business analysts with no ML experience. They need results within a few days.

Which Google Cloud service should you recommend?

A. Train a custom model on Vertex AI with TensorFlow

B. Use the Cloud Natural Language API

C. Build a model with AutoML Natural Language

D. Use BigQuery ML with a text classification model

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Questions Breakdown

A data science team has millions of customer records in BigQuery. They need to build a churn prediction model. The team is proficient in SQL but has limited ML experience. They want to minimize data movement and deploy the model quickly.

Which approach should you recommend?

A. Export data to Cloud Storage and train with Vertex AI custom training

B. Use Vertex AI AutoML Tables

C. Use BigQuery ML to create a logistic regression model

D. Deploy a pre-trained model from Vertex AI Model Garden

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Questions Breakdown

A company runs a Windows-based enterprise application with per-processor Oracle database licenses. Compliance requires the application to run on physically dedicated hardware that is not shared with other tenants. The workload is steady and predictable.

Which compute configuration should you recommend?

A. Compute Engine N2 instances with committed use discounts

B. Sole-tenant nodes with committed use discounts

C. GKE Standard cluster with Windows Server node pools

D. Compute Engine Spot VMs to minimize costs

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Questions Breakdown

Your company has decided to migrate from an on-premises infrastructure to Google Cloud. The workloads to migrate include a web server on Kubernetes, Apache Hadoop jobs, and a Network-Attached Storage (NAS).

What combination of services would you use on GCP?

- A.** Migrate Kubernetes application to Google Kubernetes Engine (GKE), Apache Hadoop to Cloud Dataproc, and NAS to Cloud Filestore
- B.** Migrate Kubernetes application to Cloud Run, Apache Hadoop to Cloud Dataproc, and NAS to Cloud Storage
- C.** Migrate Kubernetes application to Compute Engine VMs, Apache Hadoop to Cloud Dataflow, and NAS to Cloud Filestore
- D.** Migrate Kubernetes application to Anthos, Apache Hadoop to Cloud Dataproc, and NAS to Cloud Storage

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Questions Breakdown

A startup is deploying a new container-based, stateless microservices application. Traffic is unpredictable, ranging from zero to sometimes thousands of requests per second. The team wants to minimize operational overhead and pay only when the application is serving traffic.

Which compute platform should you recommend?

A. Compute Engine with a Managed Instance Group

B. GKE Standard

C. Cloud Run

D. App Engine Flexible Environment

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Day 1 Wrap-Up

Tomorrow's Agenda:

- Configuring network, storage, and compute
- Designing for observability, security, and compliance
- Deployment and operational best practices
- Case studies solutioning